

Continuous Capture of mAbs—Points to Consider and Case Studies

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28.1 INTRODUCTION

Continuous processing is one of the smartest and most beautiful concepts ever invented to produce more of a product to meet very fast growing demands, to do so at lower cost in foresight of reduced price levels in a “mass market,” and to do so at a higher, much more reproducible quality.

Despite of the certain fact that classic “manufacture” has its own beauty as well (more that of an art), our modern life would look very different, likely less pleasant and less affordable, without continuous manufacturing in many industries supplying materials and finished goods to us. [Table 28.1](#) illustrates the spread of continuous processing for a number of fundamentally important goods we consume or use on a daily basis.

Any methodology in a manufacturing environment ought to have a specific purpose, and there should be well-understood reasons to prefer one over another. Mass production of low margin goods is the common denominator of the methods listed in the table. Multiple times higher productivity was achieved when comparing the continuous method with its noncontinuous predecessors. For example, Ford could produce its cars in eightfold shorter time with fewer people and less space used in manufacturing. The purpose for the introduction of continuous processing was crystal clear, even though the path to their introduction was not always quick and straightforward. Blast furnaces, for example, reached Europe only about 1000 years after their introduction in China.

Quantities of biopharma products, even those with highest demand are microscopically small (see [Chapter 4](#)) relative to steel, glass, oil, or finished goods such as cars. Despite of that, some of the products listed in [Fig. 28.1](#) are produced with continuous cell culture in low kg quantities for global demand. It would be unusual to claim that those biopharmaceuticals are low-margin products (see the discussion in [Chapters 2 and 3](#)). Two of the major drivers for the introduction of continuous processing do not really apply to the biopharma industry. However, the opportunity to lower the production cost of biotherapeutic products, which are considered very expensive to make (tens to hundreds of dollars per gram of bulk drug substance, see [Chapter 55](#)) definitely motivates the study of continuous manufacturing, even in low quantity businesses with high margin. Konstantinov and Cooney [1] review a large variety of aspects to be aware of in this context, and continuous manufacturing has turned into a major topic of interest at scientific meetings around the industry.

A common feature of a continuous process is its superior productivity. Even without any detailed evaluation, the speed and elegance at which many of the processes in [Table 28.1](#) seem to effortlessly produce their products suggests that the continuous flow of material through the process must be the most efficient and economical way to run it. Nothing ever stops, and all parts of the installation are always “busy” and well utilized.

However, as [Fig. 28.2](#) illustrates, for a process to be called continuous, steady state is not a requirement, not all steps need to be operated in a continuous fashion, and the duration of the process may be finite. In the bioprocess example, the fully synthesized product, once removed from the bioreactor, is being purified in a series of batch operations through which it moves continuously. The first purification step (capture) can be run as so-called periodic counter-current chromatography (PCC), with continuous loading of columns in the line. This is the technology that the chapter will discuss in more detail. In subsequent steps, the product may still move from step to step without hold (see case study later in this chapter), but each of them may operate as a batch method. PCC for capture chromatography of mAbs can be used together with continuous perfusion culture [2] or fed-batch cell culture, as well as with subsequent steps continuously operated or as batch operations. Continuously operated virus inactivation reactors have been reported as “under development” [3]. Continuous polishing steps in the purification sequence have been discussed by several authors [4–6].

TABLE 28.1 Continuous Processes From Other Industries: Mass Production of Low Margin Products

Product	Method	Comments
Iron	Blast furnace	China since AD 1st century
Steel	Continuous casting	Since 1950s
Paper	Fourdrinier machine	Around 1800, United Kingdom and United States
Oil	Continuous distillation	Petroleum refineries
Cement	Rotary kiln	Around 1900, United Kingdom
Glass	Float glass process	From 1960s, United Kingdom
Synthetic fibers	Melt-spinning process	1950s–70s, United States
Electricity	Power plants	Since 1870–80
Water	Wastewater treatment	Introduced from 1850
Cars	Assembly lines	Ford ^a , 1908–13

^aHenry Ford was the sponsor, not the inventor, of the assembly lines developed at his factory.

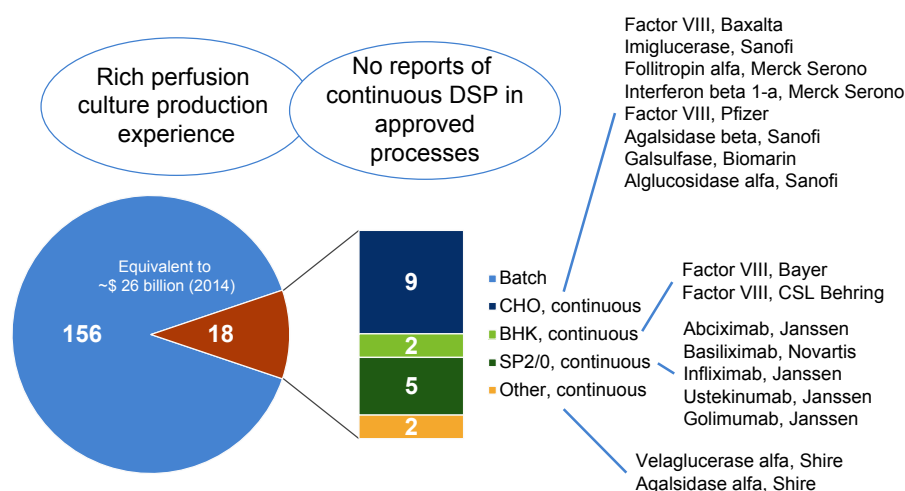


FIG. 28.1 Use of continuous perfusion culture in biopharmaceutical protein manufacturing. Data from EMA Public Assessment Reports (EMA website accessed January 2016).

A car assembly line serves as an example of a different continuous process: here all steps are small incremental batch operations performed at distinct work stations. The product moves continuously and, in contrast to the bioprocess, gets “synthesized” (assembled) while passing the different stations. The definition of what is considered a continuous process remains rather pragmatic in this way and can be applied as long as the product is moving through without major stops or bottlenecks.

28.2 THE RATIONALE FOR CONTINUOUS PROCESSING IN BIOPHARMA

28.2.1 Upstream Process

In fact, perfusion culture has been used in a significant number of cases since the early days of the biopharma industry (Fig. 28.1, compare Chapter 58). Among those are blockbuster mAbs such as Infliximab, Ustekinumab, and Golimumab, but most are coagulation factors and replacement therapy enzymes. The reasons for using perfusion culture vary for these molecules: while coagulation factor VIII has limited stability in the cell culture environment and clearly benefits from being removed from the reactor within a relatively short time after synthesis [7], the benefits for the other molecules are less well

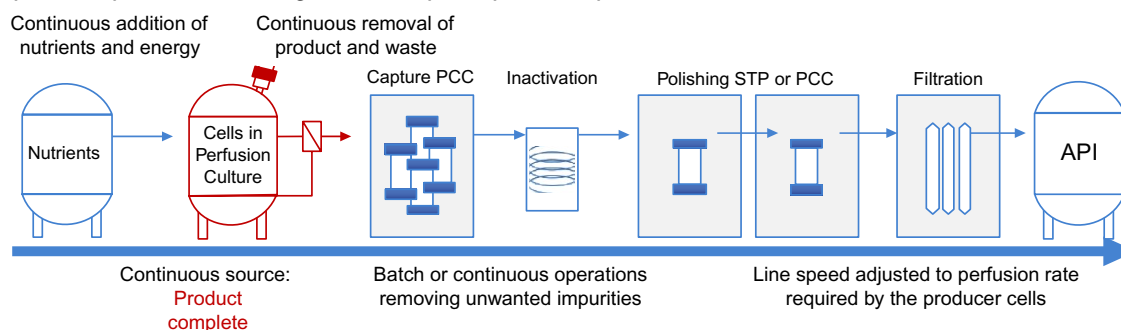
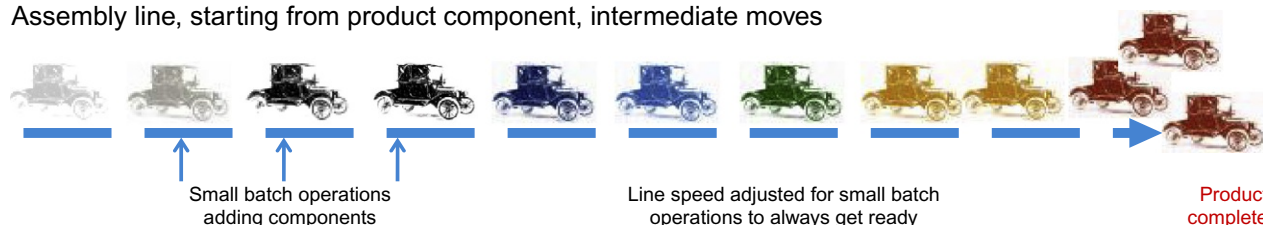
Biopharma process, starting from complete product, product moves**Assembly line, starting from product component, intermediate moves**

FIG. 28.2 Continuous production processes. Typical bioprocess with continuous perfusion culture (steady-state), continuous purification in the capture step (periodic counter-current chromatography, PCC), and a series of batch purification operations through which the product moves continuously (straight-through processing, STP). The product (API, active pharmaceutical ingredient) is completely synthesized already from the beginning and moves through the line while impurities are being removed. By comparison, a car assembly line involves a large number of small batch operations which subsequently add components. The product “synthesis” is complete only at the very end. This process allows the production of different versions of the product in the same flow.

documented. Stability in cell culture conditions could be one reason; toxicity to the producer cell could be another. Low cell line productivity could be a third reason to go continuous.

One definite common denominator of perfusion processes is the possibility to use a facility with significantly smaller footprint (see [Chapter 4](#)), a strong contributor to reduced cost of manufacturing facilities ([Chapter 55](#)). This is owed to the higher volumetric productivity of continuous perfusion culture; for example, even a culture achieving only 1 g/L/day would produce almost 30 kg (2000 L reactor) in the typical time for a batch reactor to complete a run. In that time, a current typical batch yield would only be 10 kg from 5 g/L of product titer. In other words, the reactor could be 3 times smaller for the same product output. This ratio could be further improved if the perfusion culture would be optimized to higher productivity. The experience to go down this route definitely exists in the industry, as confirmed by marketed products using perfusion technology ([Fig. 28.1](#)).

28.2.2 Purification Process

In recent years, there is a stream of publications confirming that companies and academic groups are indeed evaluating and possibly beginning to implement various options of continuous purification technology [5,6,8–14]. Proof statements and other documentation are also provided by technology suppliers [4,15–24]. In most of these published studies, the scope is limited to the capture step in mAb purification, typically performed with Protein A affinity chromatography (see [Chapter 17](#)). However, there are also papers looking at integrating the whole process into one continuous operation [4,5,25]. Products other than monoclonal antibodies, such as adenovirus, stem cells, and a malaria vaccine candidate, have been studied with continuous purification [26–28]. A proof of concept study with two target molecules, one mAb (stable) and one recombinant enzyme (unstable), was presented by Warikoo et al. [29]. They conclude that the integration of a perfusion cell culture process with a PCC system would be even more advantageous in terms of decrease of equipment footprint and removal of nonproductive steps like clarification and intermediate hold.

The purpose statement that may be extracted in support of continuous capture operations from most papers focuses on reduction of cost for the Protein A resin and the buffers consumed during this process step. This is achieved via the productivity increase from higher load capacity and better utilization of the lifetime of the resin, that is, a reduced column size. Higher utilized binding capacity per liter of resin translates into a corresponding reduction of buffer volumes per gram of product. A number of authors agree that the cost advantages to be expected on resin consumption are most

relevant in production scenarios with low demand and low required batch number (R&D stage preparations and small demand marketed products), with low utilization of the lifetime of Protein A resin when used in batch operations [8,11,30].

The savings in resin volume that are reported range from 30% to 80%, with a consensus level around 40% [6]. An increase in concentration of eluted product vs. batch mode can be expected, together with small yield improvements dependent on the process setup. Results are not easy to compare since the conditions and assumptions are not always completely communicated or comparable; for example, the two options are not optimized to the same level of ambition. In the extreme of such mismatched scenarios, 80% resin volume reduction versus batch mode has been stated in a promotional paper [21], comparing different total processing time per batch for the two modes.

Quality improvements or quality enabling features such as the removal of sensitive products from the process environment are expected features from continuous cell culture. For continuous purification, the burden is still on the technology studies to prove any advantages on quality aspects. So far, the papers reviewed for this chapter show “similar” quality, often limited to data about removal of aggregates, host cell proteins (HCP), DNA, and ligand leakage, but not related to glycosylation patterns, for example. As long as good care is taken to maintain the performance of columns over multiple runs, even under the overload conditions practiced in continuous mode, there may be no reason to assume that the quality should vary or even deteriorate during feasible use periods of the column. Considering the reported data, however, quality improvements such as lower impurity levels cannot be expected.

In the subsequent paragraphs, we will describe continuous purification technology, focusing on its potential use in mAb capture chromatography, and selected key aspects for implementation, such as development strategies and economics, with the ambition to create a balanced view of where the benefits of continuous purification can be found and how to best realize them.

28.3 TECHNICAL OPTIONS FOR CONTINUOUS PURIFICATION

The concept of continuously operating chromatographic separations is not new. The technology has initially been named “Simulated Moving Bed chromatography (SMB)”. It was invented by Broughton et al. in the 1950s for large-scale purification of *n*-paraffin [31–34]. Schulte and Strube [35] reviewed the use at industrial scale, for example, in isocratic separation of chiral mixtures. The productivity of SMB systems for separation of enantiomers has been stated at 1500 g/day and kg of chiral stationary phase with a cost of \$100 per kg at multiton level [36], that is, scalability may be considered proven in the chemical pharmaceutical industry. The methods developed for this purpose are limited to two-fraction separations and cannot use linear gradients. In purification of biotherapeutics, highly selective affinity chromatography is the only widespread two-fraction group separation, and linear gradients are typically not employed for it. However, the method variants used in bioprocessing do enable gradient elution of the product as well.

In discussions of continuous chromatography for biologics, additional terms used for the concept are “sequential multi-column chromatography (SMCC)” or “periodic counter-current chromatography (PCC)” [33,34]. For illustration, assume a continuous feed stream containing the mixture to be purified flowing in one direction, and the chromatographic adsorbent moving in the opposite direction (“counter current”), continuously extracting the component(s) that bind strongest. Instead of truly pumping the adsorbent against the feed stream, the latter movement can be “simulated” by “periodically” switching (with the help of valves) a packed column into the path of the feed to meet the flow of the mixture, and by replacing the column with a fresh one as soon as the first is saturated to the desired degree. Compared to a batch process with one column, this approach divides the packed adsorbent bed into several identical smaller columns that all become part of an “adsorbent train” moving in the opposite direction of the feed stream (Fig. 28.5). Detailed scientific descriptions can be found elsewhere [37,38].

The principles of chromatography of biomolecules have been discussed earlier in this book (Chapter 16). It should be noted that continuous purification of biologics does not address a problem that cannot otherwise be technically solved, as one may consider it to be the case with the enantiomeric separations, and there are not really any principles applied in PCC not already known from batch chromatography (e.g., mass transport phenomena). In addition to the greater technical complexity of continuous processing equipment, this may be the major reason why there has been relatively low interest during the first two or three decades of large scale downstream processing of recombinant proteins. However, at this time, the progress that can be expected from more efficient use of hardware and consumables in a continuous process fits well into the current climate of cost focus and delivers the right motivation to consider it more seriously.

28.3.1 Utilization of Resin Capacity

Fig. 28.3A illustrates how the use of the available binding capacity of a resin in a typical batch mode column process remains incomplete for the principal reason that the product of interest cannot be bound to the maximum capacity without

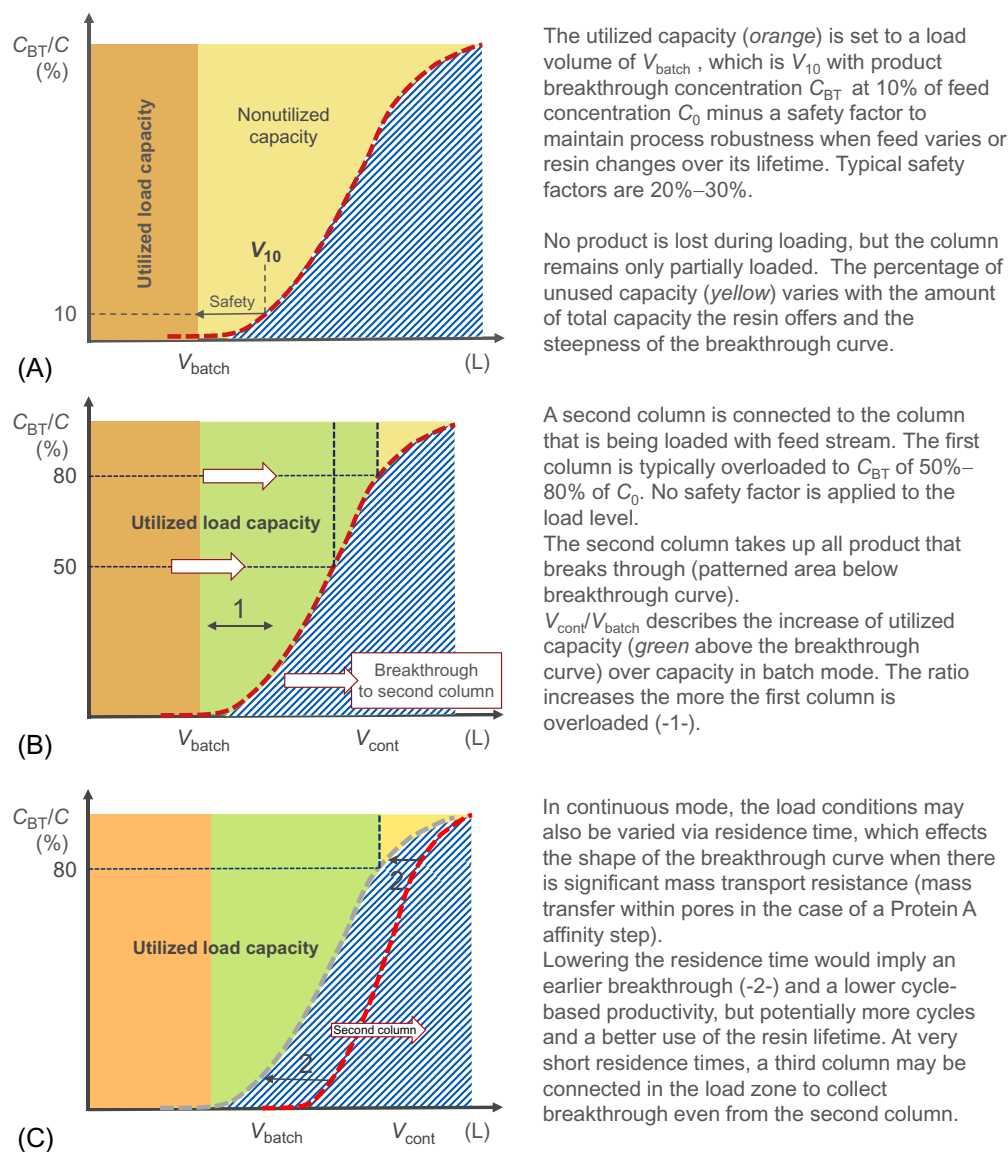


FIG. 28.3 Higher utilized capacity with continuous operation. (A) Batch mode (resin with low capacity), (B) continuous mode (resin with low capacity), and (C) continuous mode (resin with high capacity).

substantial amounts of product being washed out from the column and lost to waste. The product to be bound travels faster through the column than it can reach the binding sites throughout the porous resin structure.

In order to avoid any product losses in practice, one is not even using 100% of the load volume before breakthrough occurs, but applies a 10% reduced load relative to a load resulting in a breakthrough at 1% of C_0 [9], alternatively up to 30% down from dynamic binding capacity at 10% breakthrough. This strategy avoids the loss of excess, nonbound product in the postload wash step (PLW) and makes the batch process more robust against variability in the feed stream, in the resin between resin lots, and to loss of capacity over time as a result of repeated column cycling (safety factor).

The objective in continuous processing may be stated as “utilization of 95% or more of the resin capacity.” To overcome the inherent inefficiency of capacity utilization in batch operations, in continuous setups an identical second column is connected in series with the first column being loaded (two columns in the so-called “load zone,” see Fig. 28.5). This allows catching any product breaking through on the second column and enables loading the first column not only without a safety factor, but also beyond breakthrough (Fig. 28.3B). Typically, a breakthrough of up to 80% of the load concentration C_0 can be used to reach the objective and be allowed without risking the capability of the second column to adsorb all product breaking through. If necessary, a third column could be placed in series after the second column in the load zone,

that is, to catch breakthrough from the second column in a case where the breakthrough from the first column exceeds the dynamic binding capacity of the second column.

So far, the assumption was that the residence time for load in batch and continuous mode is kept the same. However, once breakthrough of product is not a concern, one may also run the product load faster, at lower residence time for binding. This lowers the binding capacity and the cycle-based productivity available from Protein A resins (shallower breakthrough curve) in batch mode (Fig. 28.3C). In PCC mode, the overload keeps the actual utilized capacity largely at level within the practical range of residence times and an increase of time-based productivity becomes possible. Shorter residence time can lead to higher productivity even in batch mode, but at the cost of unutilized capacity and proportionally increased buffer consumption [8].

The sources of more efficient resin use this far are:

- no load safety factor (capacity utilization up 25% from batch level)
- overload beyond breakthrough (another 30%–50% increase of load)
- shorter load time allowing better use of the column lifetime (more cycles/time).

One may expect a significantly better capacity utilization from a PCC run as compared to batch mode chromatography. This reduces the resin cost contribution per gram of produced material correspondingly. Because buffer usage is related to column volume and not bound product, even the buffer costs per gram produced product would go down, roughly by the same percentage as capacity utilization goes up. For the comparison with batch mode at a management review, however, one should make sure that the batch process has “enjoyed” its own reasonable optimization so that “apples and apples” are compared. Published improvement statements should be carefully reviewed for this aspect.

28.3.2 Parallel Execution of Load and Turnaround Cycles

There is one more significant inefficiency in batch operations: every stage of a process step has to be operated in series with a lot of idle waiting involved, that is, first load has to be completed even though the upper part of the column bed is already fully saturated, then several washes are performed before the product is eluted, and finally the column is returned to its native state, including a cleaning-in-place (CIP) step and re-equilibration in preparation for the next load cycle. In continuous operation of a chromatography step, load proceeds continuously (ideally) without waiting for these “turnaround” stages of the column operation to complete (Fig. 28.4).

With the exception of one of the possible configurations, CaptureSMB (Table 28.2), this requires one or more additional columns to be available beyond the two columns in the load zone. Fig. 28.5 illustrates a typical setup of a 4-column periodic counter-current chromatography process (4C-PCC) as one of the options available in continuous purification. All configurations have in common that the feed stream is uninterrupted and fresh columns are supplied in time to maintain continuous loading onto a capture column in what may be described as an “adsorption train” of columns [6].

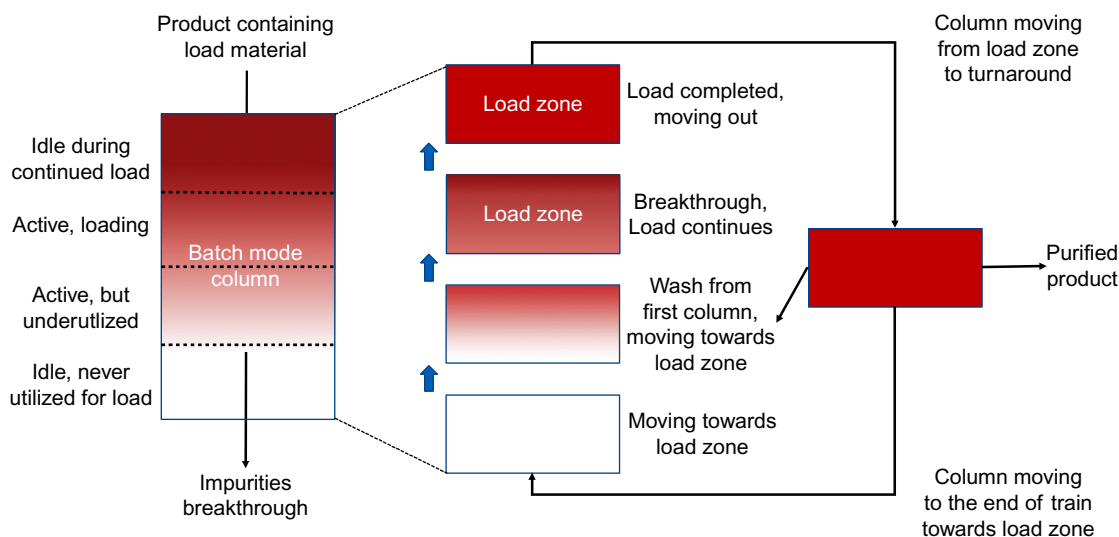


FIG. 28.4 Illustration of efficiency improvement via parallel operation, no idle parts. A continuous purification process may be illustrated by “cutting” the batch column (left) into segments (middle) that get immediately processed in the parallel turnaround cycle once they are loaded to saturation (right). Loading can continue uninterrupted to the “bottom part” of the column until each part of the column is saturated for best capacity utilization.

TABLE 28.2 Overview of Commercially Available Equipment Configurations for Continuous Mode Purification; System Suppliers Mentioned in Footnotes Own Their Corresponding Brand Names

Name	Columns	Flow Range (L/h)	Comment
ÄKTA pcc/Bioprocess pcc ^a	4	0.03–2000	Dynamic control
BioSC ^b	6	0.06–90	Predefined switching times
Contichrom CUBE ^c	2	0.01–6(540) ^c	Dynamic control
Cadence BioSMB PD/Process ^d	8(16) ^d	0.06–350	Single-use flow path, predefined switching times
Octave/SembaPro ^e	8	0.06–120	Single-use flow path ^e , predefined switching times

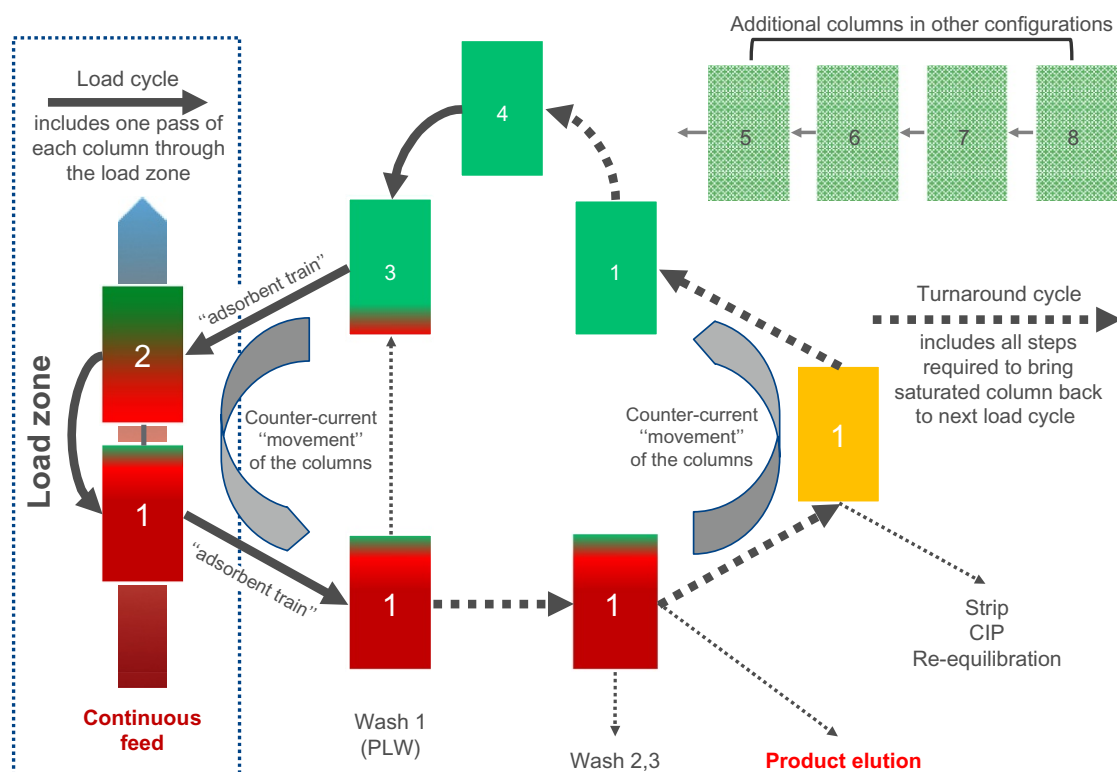
^aGE Healthcare Life Sciences, ÄKTA and customized BioProcess skids.^bNovasep, larger scale SMB systems for chemical industry available.^cChromaCon, CaptureSMB process, also marketing Lewa Ecoprime systems for larger scale.^dPall Life Sciences, the PD system allows to connect 16 columns.^eSemba, single-use flow path not yet available at time of writing this chapter.

FIG. 28.5 Illustration of a 4-column periodic counter-current chromatography system (4C-PCC) and the switching of columns between different stages of the operation. The continuous feed stream (left) is met by the simulated movement of the columns in the opposite direction (center). **Bold lines** indicate column switching in and out of the load cycle, **bold dotted lines** indicate column switching within the turnaround cycle, **thin dotted lines** indicate liquid flow in steps returning a saturated column ready for the next cycle through the “load zone.” Load is performed with two columns in connection (except for the short period of the “postload wash,” [PLW]). Washes, elution, strip, CIP, and re-equilibration happen in parallel to the continuous load, otherwise similar to a batch operation. The minimum system configurations contain either 2 or 3 columns. Other configurations may contain several additional columns (see also Table 28.2).

Two or more columns, typically two, are positioned in series in the so-called load zone. The first column is saturated to a defined breakthrough and is then switched out of the load zone where the column previously in second position moves upstream into Position 1 and begins to get loaded directly with feed as a single column in the load zone. From the saturated first column, any loosely bound product that might otherwise elute and be lost is washed in a specific postload wash step (PLW) onto the third column in the train, which afterward moves into the load zone in second position. Loading continues now as before with two columns in series, up to the target breakthrough for the column in Position 1.

The saturated first column begins additional washes (typically two), performs product elution, and is then regenerated with a strip stage, cleaning-in-place (CIP), and re-equilibration to return it to the end of the adsorption train. The whole process may be described as a “load cycle” where all columns pass through the load zone once and a “turnaround cycle” operated in parallel to the load cycle in which each column is subject to all steps required to wash out impurities, elute the product, and get the column ready for the next load. Since no impurities in fractions need to be recycled (as would be the case in enantiomer separations), constant operating conditions are reached almost immediately, that is, right after the first load cycle; from the second load cycle onward in the 4C-PCC configuration illustrated, each column gets loaded in the following sequence: first during postload wash (PLW), then as the second column is in the load zone, and finally, the first column. This means that columns entering the load zone are already preloaded with product from PLW, and when they switch to Position 1 in the load zone, also from breakthrough of the previous first column. The average or “equilibrium” load time per column is thus shorter than calculated for a clean column as in batch mode.

For continuity of the load process part, the time for the load cycle t_L needs to be longer or equal to the time for the turnaround cycle t_T .

$$t_T \leq t_L \quad (28.1)$$

In cases where t_L does not meet the condition of Eq. (28.1), the situation can be remedied by adding additional columns into the system. Eq. (28.1) represents a simple but essential constraint to respect in the development of a PCC process independent on system setup.

Apart from better capacity utilization, continuous operation thus offers an additional source of more efficient chromatographic separation:

- Decoupling load from the rest of the column process allows parallel operation of the load cycle and the column turnaround cycle.

More cycles can potentially be run in a given time, dependent on conditions chosen, than with a batch process. Both the option to load faster and to operate the whole cycle more efficiently enable the opportunity to use the lifetime of resin in the columns within a relatively short campaign and reduce resin cost per gram produced product further.

28.4 SYSTEMS FOR CONTINUOUS PURIFICATION

Systems available on the market supporting continuous purification differ in number of columns and type as well as configuration of the flow path with valves and sensors.

28.4.1 Overview of the Commercial Offering for SMB/PCC Systems in Biopharma

Table 28.2 summarizes the available offerings. All systems have in common that the load cycle can operate continuously with the turnaround cycle processed in parallel to loading.

Valve Arrangements

For lab scale development purposes (non-GMP), multiport valves can be used to build the required flow path with the benefit of flexibility as needed in development. The pilot and process scale systems apply distributed valves such as diaphragm valves or pinch valves for single-use flow paths (see Chapter 25) which are considered more feasible for GMP compliance (no rotating/sliding surfaces) despite the fact that this results in a large number of valves to be cleaned. The complexity of the flow path depends mainly on the number of columns and pumps, and the process control capabilities of the system, but in general, it is significantly more complex and requires a higher level of automation than in a batch processing skid.

Designs differ in the approach of selecting and delivering buffers to the columns: buffer selection valves can be located on the suction side of the pump as in the ÄKTA pcc and Contichrom systems. Alternatively, each buffer may get delivered by its specific pump, that is, when the buffer selection valves are located at the inlet of each column and on the pressure side of the pump, as realized in BioSC, BioSMB, and Octave systems. The former approach reduces the number of required pumps and can potentially be attached to an inline conditioning (IC) function delivering any buffer or gradient at process scale directly to the column (see Chapter 27), thereby very significantly reducing the floor space of the facility (up to 90% less buffer storage tank volume). The latter approach reduces the mixing volume and time required for buffer changes and provides for clearer segregation of buffers [39]. If other setups, for example, with combinations of PCC and subsequent steps shall be configured, additional pumps may be necessary.

Disposable Flow Path

A disposable flow path is currently offered on the Cadence BioSMB systems (PD and Process). This feature removes the need for cleaning of what is a (very) complex flow path, especially for this particular system offering 240 valves in the PD system [40] as compared to 9 valves in a 4C-PCC system (see Fig. 28.6), and facilitates process design and validation to a degree (cleaning and cleaning validation). As always with disposable technology, however, other complications such as extractables and leachables studies (E&L) and supply chain management should be considered in relation to the saving of cost and time for cleaning development and validation. Due to the complexity of the flow path, it remains a fixed design even in disposable format. The driver towards disposable technology in continuous purification would likely be the risk management aspect, that is, the wish to start each batch with a specified clean flow path, almost certainly not the direct cost aspect. There are two scenarios where single-use technology may be considered an enabler for continuous purification and could become an advantage worth paying for: a) whenever a perfusion reactor with long run time delivers the feed and the PCC system is expected to run over that whole period as well, and b) when there is very little time to worry about the readiness of a system. However, a cost/benefit analysis should be made using a realistic estimate of the expected workload in the development and pilot facility, and on the actual need for the complexity of this specific system solution. Independent on the aspect of continuous processing itself, a disposable design would also be favored, if the process shall be run in a facility that is otherwise operating with single-use technology [41].

Footprint

In first approximation, significant footprint savings should not be expected from PCC in comparison to batch mode chromatography. The systems contain a (sometimes much) more complex flow path with additional pumps and valves as well as

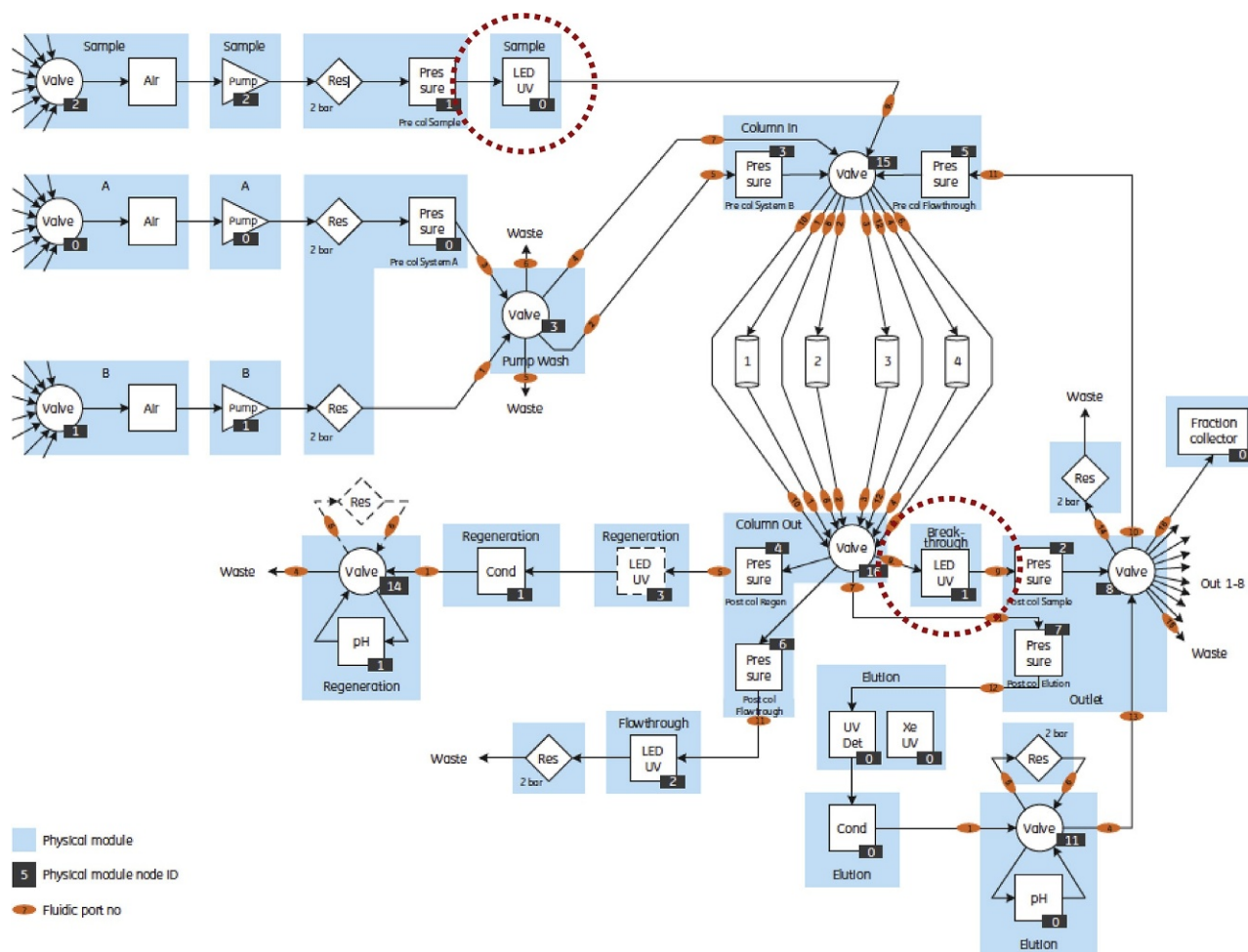


FIG. 28.6 Flow chart of a 4C-PCC system (ÄKTA pcc); feed line and column outlet UV cells (dotted circles). Image courtesy GE Healthcare.

monitoring devices and sensors. While each connected column is much smaller and even the total resin volume is smaller, arranging multiple columns with the connecting tubing likely takes more space than a single column of the larger volume required in batch, not the least because in batch mode that volume is packed to a higher bed height and thus, relatively speaking, a smaller diameter. Footprint savings in an integrated continuous process are instead a result of the smaller bioreactors operated in perfusion and subsequently smaller downstream equipment and, possibly, of the use of in-line conditioning (IC) for buffer preparation for the downstream processing skids (see [Chapter 27](#)).

28.4.2 Aspects of Column Selection for PCC or SMB Systems

Prepacked Columns

Prepacked columns are often mentioned in conjunction with the smaller column volumes in PCC processes and even more so in context with a disposable flow path on such system [40]. They remove the need to pack the multiple columns to be attached to a PCC system. While this may be considered by some to be a questionable advantage for a single-column batch scenario considering the fast and reproducible packing possible with a modern column (see [Chapter 26](#)), the multiplied time saving in a PCC scenario is more likely to make the difference in favor of prepacked columns. In the experience of the author, the thought of re-use regularly becomes a consideration for everything that is relatively expensive. This has been a discussion ever since the beginning of prepacked column availability for bioprocessing (“How often can I use them?” [42]). The question of re-using prepacked, disposable columns for several campaigns, in combination with the high number of “re-uses” with intermittent CIP during frequent cycling in one PCC operated batch, raises the issue of cleaning and cleaning validation for a product that is designed to be discarded after use. In low demand scenarios, these issues drive the user towards PCC process designs commonly favor a high number of cycles to use up the column lifetime as quickly as possible.

Column Packing Quality

An essential factor for a reliable continuous process is the robustness of quality of the columns used and, more specifically, the similarity or even identity between columns. If the columns are not identical, the theoretical calculations of parameters such as switching times may not prove to be robust during a run, and consequently, it would become difficult to design an efficient and robust continuous chromatography process. However, the packing of a column with a chromatography resin is strongly dependent on the resin and the specific packing method developed for each resin-column pair [43–45]. Even small differences in the number of plates or other packing properties can have a significant effect on the robustness of column performance over time, for example, relative to the stability of the packing [46] or determination of robust switch times. In this context, it should be noted that prepacked column suppliers may offer any resin in any of their columns, but they are unlikely to have access to the packing methods and the specific packing equipment developed for several of the modern resins considered for PCC, that is, unless both column and resin come from their coordinated development of the unit operation. Generic prepacked column suppliers are using generic packing methods in their own columns with other suppliers' resins. It is in the nature of the columns used here (disposability) that their technical features to support advanced packing methods is limited. Developing a robust packing method for a resin-column pair is time consuming and costly, that is, if a company does not have access to ample resin supply from their own sources just for the packing development [43–45,47]. Verification of these aspects should be performed as part of the column and resin selection [48]. One does not escape this challenge with continuous purification, on the contrary it may be more important than with classic batch mode operations. Technology that is not yet widely accepted may be expected to be extra thorough on risk evaluation. Absence of detailed packing studies including repeated-use studies may indicate an increased risk and would have an impact also on the selection of a control strategy for the system (see next paragraph).

Column Format—Replacement Cost

A number of other features of a PCC process are impacted by the column selection, and also vice versa: column selection impacts for example the opportunity for cost savings. There is no question that classic chromatography columns are quite a bit more expensive than the disposable column hardware used in prepacked columns. However, the convenience of prepacked columns comes at a price. Per liter of resin, this price is significantly higher compared to bulk material; in fact, the hardware cost contribution is higher for smaller columns than for larger ones (high replacement cost). The argument of cost in favor of PCC needs to be amended with a careful analysis of those “format costs” that occur with each replacement. For example, on average a liter of a typical Protein A resin in bulk would cost $\approx 50\%$ of what it costs in an 8-cm diameter prepacked column (compression factor between bulk and prepacked included in calculation). When going to 5-cm bed height, this relation would shift even more in favor of the bulk material, since the price of disposable hardware does not change significantly with changes in bed height.

Ideal vs. Available Column Dimensions

Assuming a 2000L bioreactor with 5 g/L product titer shall be processed on the capture step with a constraint on max processing time of 24 h (≈ 2200 L feed volume at a load concentration of 4.3 g/L; 10% dilution and 5% yield loss in the recovery steps), and assuming a 4-column PCC system using a typical Protein A resin at 50 g/L load capacity with 2.4 min. residence time processing the volume in ≈ 12 cycles (46 column loads), the load can be calculated to 11.6 c.v.¹ on columns that ideally should have 4.13 L packed volume. Columns, however, come in distinct diameters, and (at least for preppacked formats) also in distinct bed heights. The available diameter increments typically imply a 50%–150% increase in bed volume at constant bed height. In this case, the nearest off-the-shelf configurations would have ≈ 5 L bed volume at 25- or 35-cm diameter for 10- or 5-cm bed height, respectively, that is, they would contain 20% more resin than required in the case study. This is true unless the columns are customized for lower bed height (unlikely with a 5-cm bed height format) or unless the processing time constraint is waived. The available column reality may disturb your cost saving result, which models tend to calculate from “ideal” resin volumes. It should be noted, that the short and wide “pancake formats” typically required in large scale PCC do not make the packing easier or more robust; rather, the opposite is true. There is little experience with such aspect ratios in large scale column packing and the expectation that results from short lab scale columns are always scalable without constraints may be misleading.

When looking at published case studies, one needs to check carefully whether these aspects are considered and whether any constraints, for example, on max available processing time, max flow at pressure specification of resin and column, or flow and pressure specifications of the processing skid are being met without significant additional cost.

28.4.3 PAT Approach to Process Control in Periodic Counter-Current Chromatography

The challenges related to columns as discussed before put a demand on process control to compensate for variation in column quality, both across the group of columns connected at any time and across the lifetime of each individual column.

Systems without dynamic control require the predefined setting of column switching times from development experiments supported by vendor-supplied “development tools” helping with the calculation that puts the switch time at or near the desired breakthrough level, reproducibly, for each load on each column. However, if that breakthrough level occurs before or after the defined switch time, for example, because of variations in the feed stream or in the column capacity, the corresponding column will be over- or underloaded, respectively (Fig. 28.7).

Such variability is to be expected in biologicals manufacturing and needs to be compensated for. This calls for the use of a “Process Analytical Technology (PAT)” approach [49] with sensors that can detect the deviation and bring the process back into the specification range.

Without such control, the “column periodicity” required in continuous chromatography (switch times) and potentially the product yield could be compromised [2]. An in-line, real-time measurement of antibody in solution is currently not available. However, as long as there is a reliable way to correlate the antibody concentration with the signal of a detector,

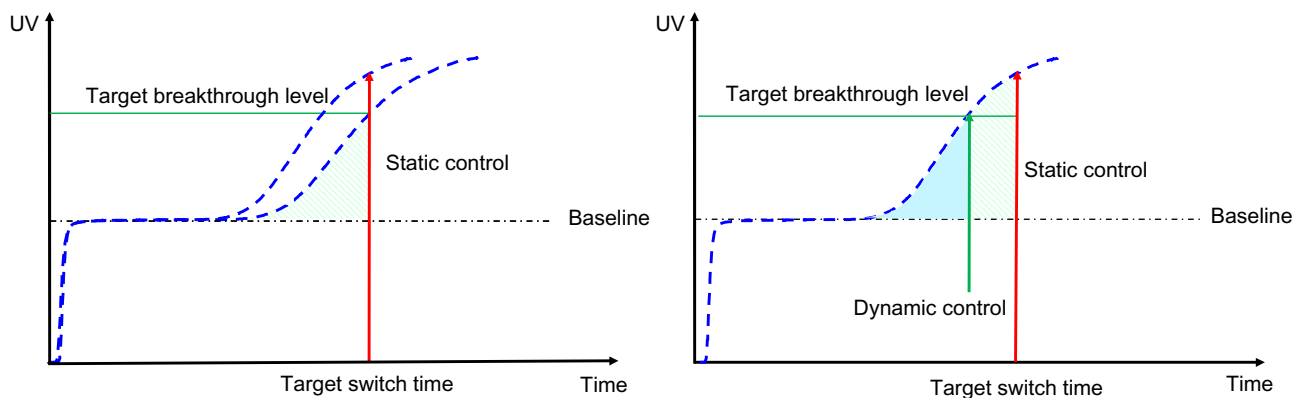


FIG. 28.7 Predefined switch times: risk for loss of control upon variability in feed stream concentration or in column performance; earlier breakthrough and overload of column (left); compensation of earlier breakthrough with dynamic control switching at the desired breakthrough level, not the preset time (right).

¹ c.v. = column volumes (dimensionless).

this lack of technology would not stop successful application of a PAT approach. The typical “protein detector” in chromatography processes uses UV/Vis absorption at a wavelength of 280 nm. Currently this method is “challenged” by high concentrations of product or impurities due to high titer in cell cultures, high binding capacity and eluate concentrations from columns, and ultrafiltration steps yielding product concentrations beyond 200 g/L.

Under such conditions, UV/Vis absorption produces a signal far beyond its dynamic range, that is, little or no information can be obtained about product concentration or, for example, the fine structure of an elution peak, even worse for a typical breakthrough UV-profile. To extend the dynamic range, a wavelength where the protein has a lower extinction coefficient is sometimes used. Off-line sampling is used to measure concentrations. Fig. 28.8 illustrates how, instead, a 0.4-mm, short path length UV cell can increase the information content from UV/Vis absorption at 280 nm in comparison to a classic 2-mm path. This example and additional opportunities for use of UV with high protein concentration have recently been presented by Ehring [50].

By placing one UV cell into the feed line before the column inlet and one under the column outlet (see Fig. 28.6), it becomes possible to record the differential signal (ΔUV) between the breakthrough and the signal from nonbinding background and use it to control switching of columns at predefined breakthrough levels independent on time, thus compensating for variability as illustrated in Fig. 28.7.

The proprietary principle is called “delta-UV” and is used for dynamic control in the ÄKTA and BioProcess pcc systems (Table 28.2). The signal from the feed line represents the total absorption from the mAb product and the impurity background. Since all antibody (and few other proteins) initially binds to the column, the signal from the column outlet after impurity breakthrough represents the absorption of the impurities. The differential ΔUV thus represents the antibody concentration in the breakthrough (Fig. 28.9).

Fig. 28.10A and C show how an improved UV/Vis absorption measurement with a wide dynamic range and with the use of the ΔUV approach supports the dynamic control of continuous chromatography and enables corrections of the set process when variations of feed input or capacity of installed columns occur: in one experiment (A), the system was challenged with a gradient of IgG feed concentration to simulate variation of product content in the material loaded onto the column. The ΔUV signal shows how the control approach kept the columns in the load zone for a longer time when the load concentration was lower, and vice versa. The amount of antibody eluted from each column remained constant, independent on what load concentration was used (B). In a second experiment (C), the connected columns were packed with two different resins of different capacity. Column 1 had the lower capacity resin and shows a consistently lower elution of mAb as a consequence (D).

All columns were always loaded to specification of breakthrough level. The flat top of the ΔUV peak in (C) indicates how the system held the low capacity column in the load zone once loading was complete, until a fresh column became available.

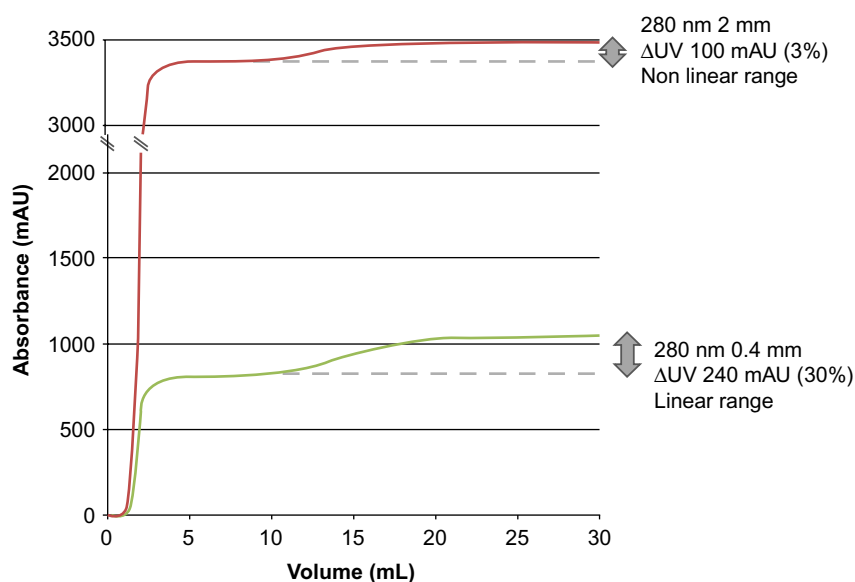


FIG. 28.8 Improved UV capabilities; 2 mm standard cell path length and 0.4 mm cell path length to handle high product concentrations and background signals.

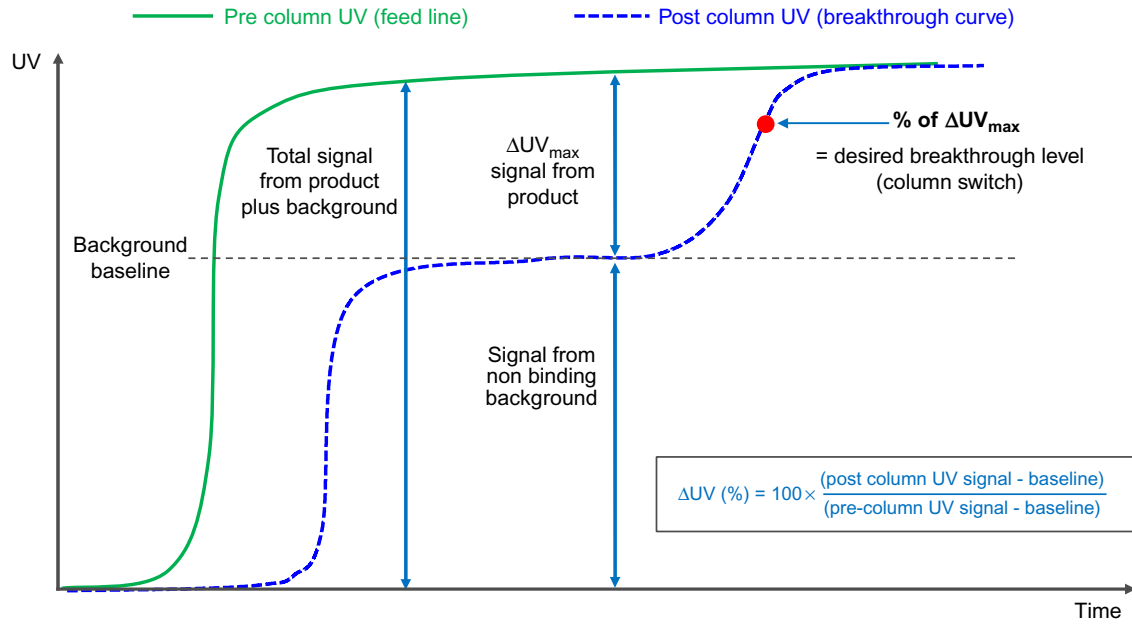


FIG. 28.9 Principle of dynamic control with the ΔUV signal [19].

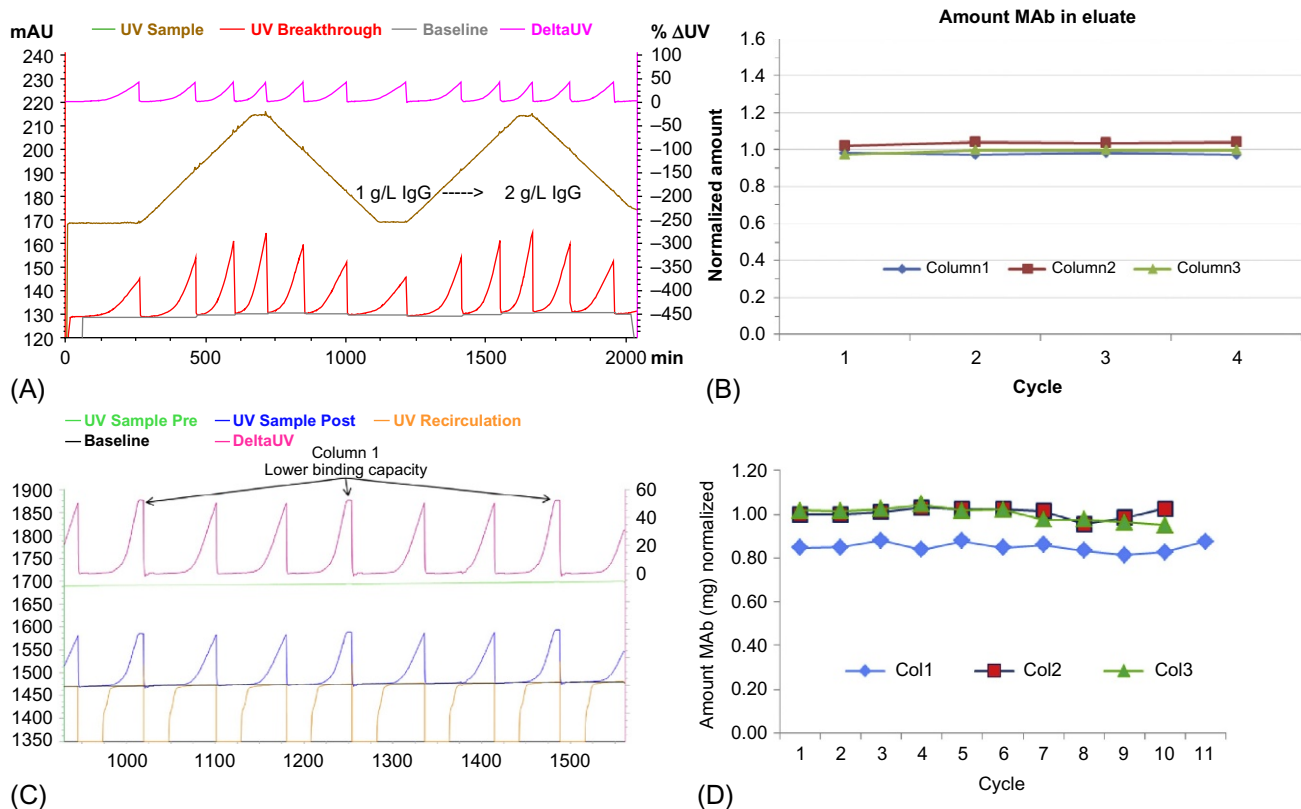


FIG. 28.10 Dynamic control with feed concentration changes (A) and column capacity changes (C) using proprietary delta-UV technology [19]; (A) simulation of variability in feed stream concentration via a mAb gradient (1–2 g/L of IgG); (B) eluted amount of antibody from experiment (A); (C) simulation of variability in column capacity via use of two different resins in the columns connected to the system; (D) eluted amount of antibody from experiment (C); ΔUV used to switch columns (pink line).

28.5 PROCESS DEVELOPMENT GUIDANCE

28.5.1 Manufacturing Scenario and Objective

At the beginning of process development, an evaluation of the production scenario to be served with the process is performed (see [Chapter 4](#)). The most markedly different candidate scenarios for PCC capture of mAbs are those supporting preclinical testing, those intended for early clinical trials and then for late clinical trials and production for regular marketing. Pollock et al. [5] provide an estimate of the product quantities required in preclinical and clinical settings ([Table 28.3](#)).

Another point in the development considerations would be an existing facility with its specific constraints, alternatively the opportunity to tailor a facility towards optimal conditions for continuous processing. Many case studies comparing batch and continuous operation assume that there is a blank sheet related to the facility and count advantages from optimized facility design as benefits from continuous processing. In other words, in such cases, it is not easy (if not impossible) to derive the exact source of a financial advantage and draw conclusions from the analysis unless there would be full transparency of all assumptions (which experience tells is very unlikely, even in academic papers).

The mAb capture step intended for regular manufacturing would be optimized to maximize utilized capacity in each cycle resulting in the maximum purified product output from the column over the entire resin lifetime (assuming the potential lifetime will be used well). This is similar to the development objective for batch mode chromatography, but the advantages of PCC in terms of capacity utilization apply. In preclinical and clinical manufacturing or other low demand scenarios, where only one or very few batches may suffice to produce the required quantity of material for testing, one aims to minimize the volume of resin required to produce the limited quantity needed by performing as many cycles as possible in the given time (knowing one will otherwise not use the potential lifetime well and cost of resin per gram of mAb may be high).

For the former scenario, one often expresses productivity as cycle-based, and for the latter, as time-based [8].

$$\text{Cycle based productivity} = \frac{\text{mass}_{\text{cycle}}}{L_{\text{resin}}} / \text{time}_{\text{load cycle}} \quad (28.2)$$

$$\text{Time based productivity} = \frac{\text{mass}_{24\text{h}}}{L_{\text{resin}}} / 24 \quad (28.3)$$

We discuss selected examples of these development options in the last paragraph of this chapter and compare with batch mode alternatives.

28.5.2 Key Information to be Developed for PCC

One of the concerns expressed about PCC is related to the relative complexity of the approach, and consequently, the expectation that the development of such processes may be complex and time consuming. PCC is not that much different from a batch mode chromatography process though: the same resins are being used in the same columns, and at least for most of the systems, the hardware and the process control approaches are similar to a batch mode skid. Purification with PCC can be understood as illustrated in [Figs. 28.3–28.5](#): instead of in one column, the same loading process happens sequentially in

TABLE 28.3 Product Quantity Estimates for Preclinical And Clinical Testing [5] and Commercial Manufacturing

Testing Phase	Product Quantity (kg)	Comment
Preclinical	0.5	Single batch
Phase I and II	4.0	Single batch
Phase III	40.0	Multiple batches dependent on scale and titer
Commercial	10–2000	Single or multiple batches dependent on scale and titer

Based on author's own data, see [Chapter 4](#).

several columns that replace each other at a rate defined by the time it takes to saturate a column and the time it takes to get a column ready for the next load. The latter can be shortened by putting additional columns into the parallel turnaround cycle, that is, in cases where load is faster than turnaround. The balance between load and turnaround time, the “feed continuity” requirement, is the key constraint in the system different from batch mode, where load is not happening continuously and the steps for turnaround are performed in sequence, not in parallel with the load step. The other constraint is caused by the wish to operate the process at short residence times potentially leading to pressure flow limitations, especially while two columns are connected during load.

The Load Process

In principle, the ΔUV control approach described earlier allows to stop the load process when there is no fresh column available and to complete the load process while holding a fresh column that is already available, and to do so even when there is variability in the feed or column capacity. Robustness of the purification process is achieved and yield loss is avoided via dynamic control (ΔUV). However, both situations imply a compromise on the productivity of PCC, since parts of the system would remain idle during those hold times, that is, feed continuity or productivity targets may still be violated. This control approach thus addresses what likely is one of only two reasonable robustness concerns in a management review. The other is the risk for infection/contamination of the system during very long periods of operation as may occur in scenarios where the feed is from a perfusion culture. In the vast majority of cases, in fed batch culture scenarios, the processing time would not be longer than for batch mode purification, that is, at maximum defined by the proven stability of the feed stream without processing or more often by practical constraints such as work shift regimes, that is, 8–24 h. Unless the flow path of the PCC system is very complex (or does not use distributed single valves), this represents a challenge that has successfully been addressed in manufacturing suites hundreds if not thousands of times, usually via stringent CIP protocols and good manufacturing practice, that is, not with enhanced measures like steaming-in-place (SIP), and not only with single-use technology.

What needs to be developed to minimize the challenge with feed continuity and to maximize productivity under full control via ΔUV is the understanding of load/overload dependent on residence time, that is, the breakthrough curves at different residence times, load concentrations, and bed heights need to be experimentally generated, in principle just as with batch mode development. An example of the experimental setup is provided in [8]. Specific for PCC, one will then be able to understand how much product breaks through and gets adsorbed on the second column in the load zone dependent on the set processing conditions. This amount will increase the more the first column is overloaded, that is, the second column takes more product both when the allowed breakthrough is set higher and when the breakthrough happens earlier due to reduced contact time. Another constraint appears: as long as there are no more than two columns in the load zone, load to the first column should only continue until just before breakthrough on the second column, which sets the shortest feasible residence time (if not set by the pressure-flow constraint). The amount of breakthrough onto the second column also defines how much product already is loaded when that column switches into position one in the load zone. In addition, the column has been loaded even earlier with product from the “postload wash” before entering the load zone. The worst-case product quantity from that wash is equivalent to one column volume of load at load concentration, that is, very low when loading from a low titer culture (perfusion or legacy batch processes) or up to about 10% of the total load, for example, assuming a current 5 g/L load concentration and 50 g/L resin load capacity. Mahajan et al. [9] reported a decline in yield from 98% to 82% when the postload wash could not be loaded onto one of the columns in the turnaround cycle. For very high load concentrations, for example, from a concentrated fed batch process, this percentage could be more substantial. The time for each column in position one of the load zone thus needs to be adjusted for these effects relative to an initial load time (t_L) calculation as performed for a batch column (or for the fresh columns in the very first cycle of PCC when no column has previously been loaded), which is:

$$t_L = C_L / C_F \times t_R \quad (28.4)$$

where C_L is the load capacity for fresh resin at the set residence time t_R for the load step and C_F is the product concentration in the feed material. The breakthrough going onto the second column can be calculated as the area under the breakthrough curve between the start of breakthrough and the level set as the desired load level for each column (*green* patterned area in Fig. 28.3). The area above the breakthrough curve between the set load level and 100% breakthrough represents the unused capacity of the resin (*yellow* area in Fig. 28.3), which shall be kept as small as feasible considering all constraints in the process. If one choses to create a mathematical model fitting the experimental curves, that is, to derive breakthrough behavior from the process parameters, these areas can easily be calculated and used for process optimization [8,12].

The Turnaround Process

The turnaround process time (t_T) simply is the sum of all steps after load that bring the column back, ready for the next load cycle: all washes (t_W), elution (t_{El}), strip (t_{St}), CIP (t_{CIP}), and re-equilibration (t_{Eq}).

$$t_T = t_W + t_{El} + t_{St} + t_{CIP} + t_{Eq} \quad (28.5)$$

Provided that there is more than one column available in the turnaround cycle, this time can be shortened. If the total number of columns in the system is N and N_L is the number of columns in the load zone (typically two), the limiting turnaround time is calculated from Eq. (28.5) by dividing the sum of step times with $(N - N_L)$. This optimization would be possible with a total number of columns of at least four.

In addition to increasing the number of columns in the system, which increases complexity, one may look at the individual steps in the turnaround cycle and study the opportunity to speed them up. Residence times for these steps may, in principle, be set to the pressure limit of the resin at the selected bed height (Fig. 28.11), which would represent the lowest residence time allowable for the resin-column pair. Resins with a larger bead size tend to be at an advantage when it comes to the pressure limit constraint.

At shorter bed heights, the allowed flow rate will be higher, the residence time shorter (Fig. 28.11). For the load zone where columns are connected, a lower flow rate applies (double or longer bed height). Process development needs to verify that shortening the turnaround cycle will still deliver a fresh column for load at specification, and that speeding up elution does not change product quality nor yield.

Shorter residence times may affect wash, strip, CIP, and re-equilibration specifications and as part of the verification it should be checked whether different (larger) buffer volumes (cv) than in batch recipes would be required to achieve specifications. The CIP step, typically performed with low to medium concentrations of NaOH is time controlled, that is, for the CIP agent to work as intended a certain time may have to pass (see Chapter 33). It can be assumed that the challenge with this is smaller, if good care has been taken to clarify the feed prior to the capture step, or even to reduce potentially critical impurities such as DNA in the product recovery steps. The ability of the resin to tolerate more harsh CIP conditions will also mitigate the risk when speeding up the CIP step and significantly shorter CIP times than the typical 15–30 min in batch mode have been reported in a proof of concept study for PCC [11].

Step residence times and volumes (cv) in the turnaround cycle are sometimes recommended by resin vendors for batch processes and may be used as a starting point for the optimization. Once the load cycle with its conditions has been defined and the column bed height has been decided, residence times can be adjusted to match the pressure limit followed by

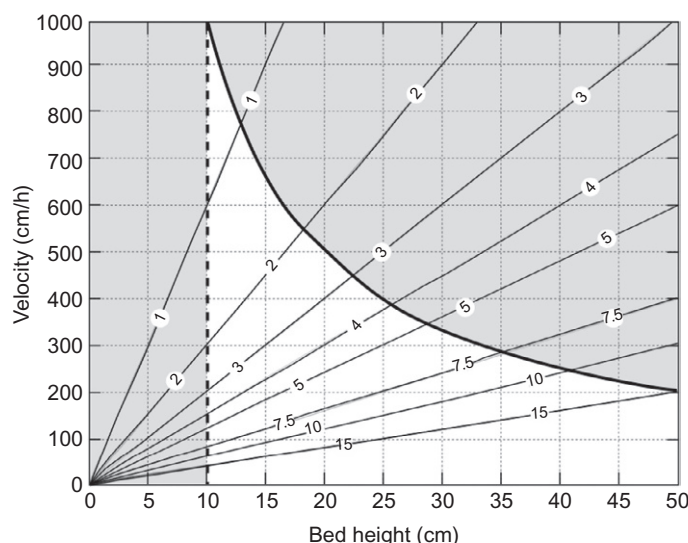


FIG. 28.11 Pressure limit, bed heights and residence times for a typical Protein A capture resin. The white area indicates the operating space for the resin assuming that bed heights below 10 cm are leading to aspect ratios in production columns that are not recommended due to the robustness of tested packing procedures (Using lower bed heights puts the user outside the resin manufacturer's operating recommendation. However, as long as user-verified packing methods are applied, a new customized specification can be defined and be acceptable to regulatory bodies.); if this resin would be packed to 10-cm bed height and used in PCC, the max load velocity would be 500 cm/h, representing a residence time of 2.4 min, but the turnaround cycle could possibly be operated at 1000 cm/h, just based on the pressure limit. Image courtesy GE Healthcare.

verification that the process still performs as expected. Dependent on which development scenario one aims for (maximized capacity utilization or resin lifetime utilization), a number of iterations of parameters discussed above will lead to a feasible sweet spot for capacity and productivity [8]. As long as the process is set to work in the pressure drop limited zone and not at the feed continuity (turnaround time) limited zone, one may expect a wide degree of freedom to adjust target breakthrough. Once turnaround time limited, the load flow rate would have to be adjusted to comply with the feed continuity constraint.

With these general guidelines, development of a PCC process is not significantly more complex than for a batch process. The challenge may be to understand the opportunities with PCC correctly and to incorporate the constraints appropriately. A robust control strategy such as ΔUV mitigates the risk to acceptable levels and holds the promise to simplify development significantly; at the Recovery of Biological Products XV conference 2014, Karol Łacki proposed an approach for the development of both batch and continuous mode chromatography [51] to address the question whether a switch from batch to continuous mode chromatography or vice versa would be feasible and could be supported by process characterization data generated based on a unified development strategy. Similarities and differences between batch and PCC mode have already been discussed previously: how can one minimize the number of experiments leading to all necessary information characterizing both modes of operation?

The suggested approach attempts to “look into the batch column,” by “breaking it up” into few smaller columns/segments, for example, four as in a 4C-PCC setup as illustrated in Fig. 28.4. Placing a UV cell at the outlet of each segment allows that “look into the bed” and thus monitoring of breakthrough curves after each of the smaller columns. With this approach, breakthrough curves for four different bed heights/residence times can be generated in a single experiment by employing the ΔUV approach (see Fig. 28.12). Thus, determination of dynamic binding capacity at various breakthrough levels as a function of residence time becomes possible. Also, the effect on wash and elution pool volumes as a function of the level of resin saturation can be extracted from one experiment. Consequently, very few experiments will be needed to describe the operating space and the essential characteristics of the PCC operation and at the same time the batch mode purification.

This approach supports a simplified characterization of PCC processes and ultimately their validation. If loaded from a fed batch cell culture, batch definition is as simple as for batch mode chromatography: one batch of eluted product accumulates from the cycles run to process it. If loaded from a perfusion culture batch definition may relate to pooling of material eluted from PCC cycles or simply on a daily perfusion volume, for example.

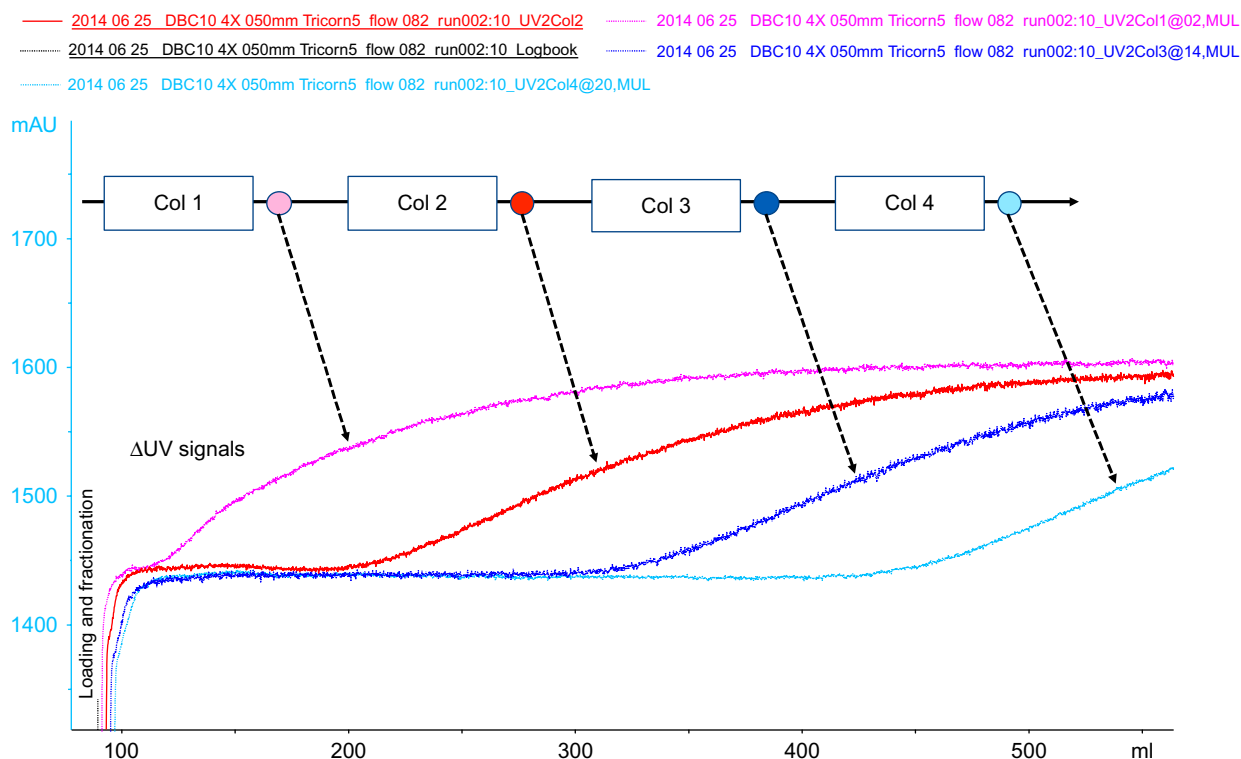


FIG. 28.12 Unified experimental setup to characterize both a batch and a PCC mode capture process at the same time [51]. Using ΔUV to measure breakthrough after each column segment, one linear flow rate through the column arrangement, different residence time at each UV cell (circles). Image courtesy GE Healthcare.

An approach to determine the conditions for PCC for systems without ΔUV is described in [12,18,52]. Process robustness remains dependent on exact setting of switch times.

28.6 CASE STUDIES—CAPTURE AND POLISHING

In the following paragraph, we summarize the results of two case studies performed at GE Healthcare R&D. First, the purpose was to provide experimental proof of the key statement about PCC made earlier in this chapter and in many of the referenced papers: resulting in similar quality, the principle of periodic counter-current chromatography allows the column to be loaded to a higher capacity, which leads to reduced chromatography resin and buffer consumption, shorter processing times and consequently potential cost savings. Conditions and system layout were not further optimized, for example, to identify the sweet spot for savings. In the second study, we looked at connected polishing steps using “straight-through processing (STP),” which uses technology for in-line adjustment of intermediate product as described for in-line buffer delivery in [Chapter 27](#). This method itself does not apply continuous processing technology, but still provides for continuous flow of the product through the process.

28.6.1 Capture—PCC to Increase Chromatography Media Capacity Utilization

A continuous capture step for a monoclonal antibody was performed using three Protein A columns connected to a periodic counter current (3C-PCC) system (see [Table 28.4](#)). The main focus for this study was to maximize the capacity utilization of the chromatography resin.

The loading strategy for the PCC runs was determined based on breakthrough studies using frontal analysis with clarified cell culture supernatant. Frontal analysis experiments were performed at different residence times. Based on the breakthrough curves the dynamic binding capacity was calculated, and residence time and breakthrough levels were evaluated to maximize capacity utilization but at the same time keep the loading time longer than the time for the turnaround steps. A residence time of 5 min and a breakthrough level of 50% were selected.

The chromatogram from the loading step with 10 cycles on the three-column setup (30 loadings) is shown in [Fig. 28.13](#). Each cycle with one run per column is marked in the chromatogram. The running conditions for the purification are described in [Table 28.5](#). The UV profiles during loading were evaluated to ensure consistent column performance over 10 cycles. The robustness of the purification can be evaluated by looking at the shape of the curves and as can be seen the UV curves have identical height, length and shape in each run.

The loading was controlled using the dynamic ΔUV principle. Another way of evaluating the consistency of such a run is to calculate the loading time (the time a column spends in the loading zone) for each column and cycle. [Table 28.6](#) shows that loading time was consistent regardless of column used.

Results for Capture Study

In these PCC runs, 67 g mAb/L chromatography resin was loaded onto the column compared to 43 g mAb/L in batch mode. Forty-three g/L corresponds to a load of 70% of QB₁₀ at 6 min' residence time, that is, including a safety factor of 30%. This increase in load gives a 56% increase in chromatography media capacity utilization.

TABLE 28.4 Materials and Methods for the Capture PCC Case Study

Sample	Monoclonal antibody (4.5 g/L) expressed in Chinese hamster ovary (CHO) cells. The MAb was produced in a fed-batch cell culture
System	Periodic counter-current system (ÄKTA pcc)
Chromatography resin	Protein A resin: MabSelect SuRe LX
Columns	HiScreen columns (CV = 4.7 mL)
Analytics—aggregates	Size exclusion chromatography on Superdex 200 increase
Analytics—HCP	anti-CHO HCP antibodies (Cygnus Technologies)

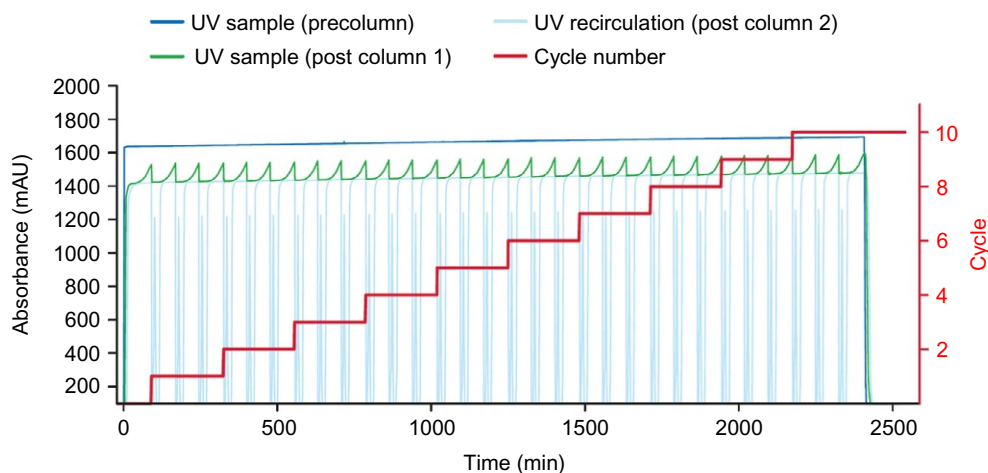


FIG. 28.13 Chromatograms (UV curves) from 10 cycles on 3 protein A columns (total of 30 loadings). Red curve shows cycle number.

TABLE 28.5 PCC Capture Chromatography Conditions

Step	Buffer	Column Volumes	Residence Time (min)
Equilibration	10 mM phosphate buffer 27 mM KCl 140 mM NaCl pH 7.4	5	3.4
Sample	Clarified cell culture supernatant. MAb concentration 4.5 g/L		5
Wash 1	10 mM phosphate 27 mM KCl 140 mM NaCl pH 7.4	1.5	6
Wash 2	10 mM phosphate 27 mM KCl 140 mM NaCl pH 7.4	3.5	3.4
Wash 3	50 mM acetate buffer pH 6	1	3.4
Elution	50 mM acetate buffer pH 3.5	4	4
Strip	100 mM acetate buffer pH 2.9	2	3.4
CIP	100 mM NaOH	3	5
Equilibration	10 mM phosphate buffer 27 mM KCl 140 mM NaCl pH 7.4	5	2

TABLE 28.6 Loading Times (min) for Each Column and Cycle

Column/Cycle	Loading Time	Column/Cycle	Loading Time	Column/Cycle	Loading Time
C1/01	92 ^a	C2/01	82 ^a	C3/01	79
C1/02	81	C2/02	79	C3/02	84
C1/03	76	C2/03	77	C3/03	80
C1/04	80	C2/04	78	C3/04	78
C1/05	81	C2/05	78	C3/05	78
C1/06	81	C2/06	78	C3/06	78
C1/07	80	C2/07	78	C3/07	78
C1/08	80	C2/08	77	C3/08	78
C1/09	80	C2/09	77	C3/09	78
C1/10	80	C2/10	77	C3/10	77

^aSteady-state is not reached.

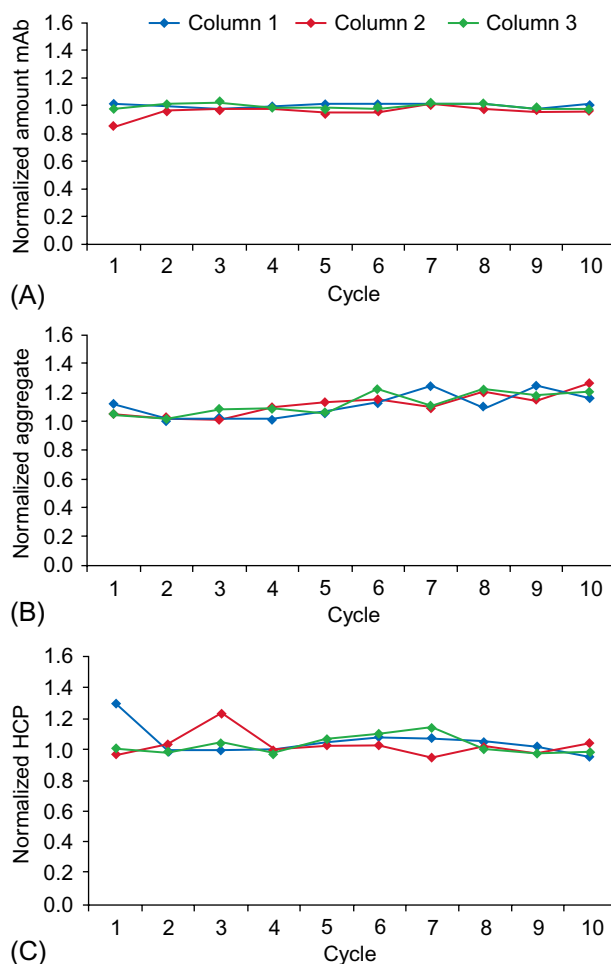


FIG. 28.14 Performance comparison in terms of (A) amount MAb, (B) aggregate content, and (C) HCP content in elution pool from the 3C PCC setup. The results are normalized against the elution pool from the first column in the second cycle (when steady state is reached). Column 1 (*blue*), column 2 (*red*), and column 3 (*green*).

The elution pool from each individual column was analyzed with respect to the amount of mAb, aggregate content and HCP content. The results are presented in Fig. 28.14 and color-coded according to column. As can be seen, the performance is robust over the 10 cycles for both purity and amount MAb in the elution pool.

28.6.2 Polishing Steps Using Straight-Through Processing

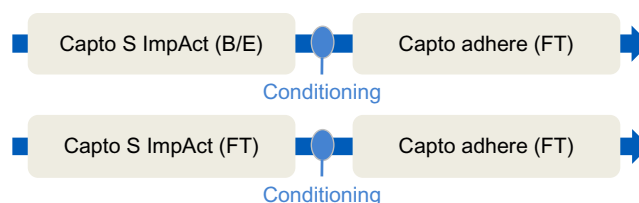
Downstream purification processes are typically performed in batch using hold-up tanks between the different steps. Each chromatography step is optimized in terms of buffer substance, pH and conductivity for binding and elution, and separate (off-line from the process flow) conditioning steps are often required to adjust the intermediate product between steps. In order to remove or minimize hold-up tanks and to avoid the conditioning as an extra operation, the concept of straight through processing can be applied. The definition of straight through processing is rather simple: two or more chromatography steps are connected in series, in-line adjustment of process streams is performed between columns to ensure correct load conditions for the next chromatography step [53]. The latter uses the same technology as the In-line Conditioning (IC) systems for buffer delivery from concentrates (Chapter 27).

Straight-through processing has been developed and tested for two polishing steps in a monoclonal antibody purification process. The monoclonal antibody was first purified on Protein A resin using PCC. Virus inactivation and conditioning of the sample before the polishing steps were performed off line. For material and methods, see Table 28.7.

Each step was first optimized individually and the best conditions for high yield and high purity were identified. After the individual studies the integration of the two steps was considered in terms of column volumes, flow rates and in-line conditioning. The monoclonal antibody had a tendency to aggregate at neutral pH creating a constraint on pH and chromatography

TABLE 28.7 Material and Methods for STP Polishing

Sample	Monoclonal antibody purified on Protein A using PCC. The aggregate content after protein A is 2%, HCP 1277 ppm and leached ligand 3 ppm
System	ÄKTA pure system. A standard system with extra UV detector and mixer, software UNICORN 6.4
Chromatography resin	Cation exchanger: Capto S ImpAct, Multimodal anion exchanger: Capto adhere
Columns	HiScreen columns, Tricorn 5 columns
Analytics—aggregates	Size exclusion chromatography on Superdex 200 increase column
Analytics—HCPAnalytics—leached ligand	anti-CHO HCP antibodies (Cygnus Technologies)Protein A ELISA kit (Repligen)

**FIG. 28.15** Setup for two polishing steps for purification of monoclonal antibody. Cation exchange on Capto S ImpAct was tested in B/E mode followed by Capto adhere in FT mode (top), and in FT mode followed by Capto adhere in FT mode (bottom).

resin that could be used during purification. Two different purification schemes were tested, see Fig. 28.15. One with the cation exchanger in bind/elute (B/E) mode followed by a multimodal anion exchanger in flow through (FT) mode. The other set-up was cation exchanger in flow through FT mode followed by multimodal anion exchanger in FT mode. The conditions used in these runs are presented in [54].

Conditioning of the intermediate from the cation exchange step was performed by adjusting the pH to 6.2 or 6.3 with NaOH in combination with dilution to lower NaCl concentration before the Capto adhere step. The elution/flow-through pool from the cation exchanger was mixed in-line 1:1 with 50 mM Phosphate 24 mM NaOH.

Results for Polishing Study

The chromatogram from the first verification run with cation exchange in B/E mode followed by in-line conditioning and multimodal chromatography in FT mode is shown in Fig. 28.16. Flow through from Column 2 was collected (marked in gray) and analyzed for yield, HCP, aggregates, and leached Protein A ligand (from the capture PCC step). Table 28.8 summarizes the results from the analysis. Cation exchange and multimodal chromatography are viewed as one unit operation with samples taken before and after the combined step. The yields in the table are cumulated yield over these two steps and

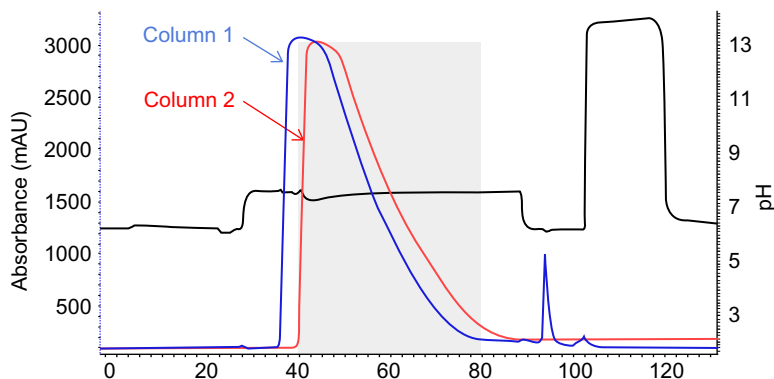
**FIG. 28.16** Chromatogram from run with Capto S ImpAct in B/E mode (blue) and Capto adhere in FT mode (red).

TABLE 28.8 Results from STP Polishing Study; Starting Level of Impurities Was 2% Aggregates, 3 ppm Protein A Leakage, and $\approx$ 1300 ppm HCP (Eluate From Protein A PCC Step)

Process Step	Yield Over Two Steps (%)	Aggregates (%)	HCP (ppm)	Leached Ligand (ppm)
Capto S ImpAct (B/E) – Capto adhere (FT)	89	0.8	16	<1
Capto S ImpAct (B/E) – Capto adhere (FT)	90	0.7	18	<1
Capto S ImpAct (FT) – Capto adhere (FT)	88	0.7	40	<1

B/E, bind-and-elute mode; FT, flow-through mode.

aggregates, HCP and leached ligand are after both steps. The process was repeated to test the robustness using a slightly higher pH for the Capto adhere step. Results were found to be similar for both runs (not shown). The yield and purity are also comparable with results from batch runs, separate for each of the methods. This study demonstrates the feasibility of connecting two process steps opening up for the opportunity to reduce footprint, eliminate the need for intermediate tanks, and increase the automation of the purification process.

28.7 SELECTED ECONOMIC CONSIDERATIONS

28.7.1 Management Review

Scientists and team leaders proposing continuous chromatography ought to be aware that most likely the “opposition” is not going to question the capability of PCC technology as such. Instead, the opposition will focus on the risks and on the business relevance of the advantages in a wider context, for example including the change of established methods and potential costs coming from the change. To meet that argument successfully, a foundation of trust needs to be the first focus: transparency and honesty in the assumptions and calculations is an absolute must. All costs need to be looked at and the scenarios should not be “tweaked” to favor the new technology. If the costs alone are not accepted as a strong-enough support argument, the proposal may become more difficult to defend, but continuous processing holds more promises worth to be considered than just direct cost savings, for example, when integrating a whole process from upstream to bulk drug substance [55,56].

Perfusion Culture

If things were just simple and straightforward, perfusion culture should be the dominating route of synthesizing biotherapeutics: consistent quality, smaller footprint, and lower capital cost for facilities should be very strong arguments. Naturally, as a consequence, the downstream line would also shrink. In a management review comparing fed batch technology and continuous perfusion culture, however, a number of other aspects may rule over the most obvious advantages (Table 28.9). Overall costs including cell culture media consumption may not be favorable, or not good enough to justify the identified risks. The space and equipment costs needed to produce large volumes of culture media supporting perfusion culture may be significant. The long processing time may be considered an increased risk for failure with difficult-to-manage

TABLE 28.9 Summary of Reasons Why New, Alternative Technology Might Not Pass a Management Review [57]

Aspect	Issue
Technology	No significant or sufficient advantage in performanceInability or uncertainty related to scale up
Quality, regulatory	Regulatory hurdles higher than for conventional methodsYielded product has insufficient quality
Economics	High cost or low yield associated with the alternative approachPoor fit with existing processes or facilities
Risk assessment	Method not well understood or appreciated (robustness)Lack of analytical methods, process controlLogistics of manufacturing: custom equipment, supply chain of consumables

consequences (a whole batch may be lost as both the time and root cause of failure is not understood immediately). Technology transfer to contract manufacturing organizations (CMOs) may be deemed difficult or impossible, a potentially unacceptable limitation to flexibility in managing the production network. The risk-averse general practice in the industry may well be behind the lack of penetration of continuous perfusion culture with biopharmaceuticals, not the least because the classic alternative, fed batch processing, is both much more established and has seen dramatic improvements in productivity over the recent years, for example, from below 1 g/L product titer towards 10 g/L or more (“concentrated fed batch”). Scale-down of the batch processes has become a very viable option. This progress reduces the potential gains from going continuous in the upstream process.

Continuous Purification

Going continuous in purification of a marketed product at this time (2017) would mean going first (Fig. 28.1). While the technology is established in the chemical industry and in some pharmaceutical processes, there is no documented case of a continuous purification process for a marketed protein drug.

While the science and engineering is beautiful and promising, the beauty of the approach needs to be translated into the language of those who focus on the money and needs to survive a robust evaluation with a large variety of criteria applied that are not just technical (Table 28.9).

In the authors' experience (confirmed in qualitative terms, in an extensive study performed by researchers at University College London (UCL) and GE Healthcare [57]), improvements that concern an isolated step in a process, that do not bring 50% or more savings for that particular step at a value of at least \$500,000–\$1,000,000 US dollars per year, are meeting implementation resistance in the decision-making process. This observation is general to any new manufacturing technology and is not limited to continuous processing alternatives. In situations where a change of an existing process is considered, the value of savings most likely needs to be higher, since the cost of change and the regulatory and supply chain “hassle” is significant (see Chapters 51 and 52). Risk assessment may overrule any cost opportunities, since revenue in biopharma is at multiples of manufacturing cost (5- to 20-fold, see Chapter 55). If a technology benefits mainly the low demand scenarios supported by R&D and a process change prior to scale up with all its extra efforts has to be envisaged, the argument of cost burden on the R&D budget (preclinical non-GMP and clinical GMP manufacturing) may prove to be a weak one. Alternatively, individual improvements are “packaged” into groups, and the savings in R&D then become substantial enough to enable new opportunities in the R&D portfolio. This will probably remain so until continuous purification has seen more widespread acceptance on the manufacturing floor.

28.7.2 Points to Consider in Financial Comparisons

When comparing two technical options for biomanufacturing where the value managed in a process batch can easily reach \$50,000,000 US dollars in revenue terms, everything needs to be on the table, and transparency needs to be created as to what aspect causes which benefit, but also which disadvantages or modification of the main advantage are present (Table 28.10).

True limitations need to be differentiated from perceived or, in the worst case, “wished-for” deficiencies in the option to be replaced (yes, there are personal preferences on both sides of the comparison, and in a creative team, there should be). Models used in such comparisons thus ought to use assumptions that represent the best performance of each option and even test potential improvements for both to see where those true limitations might be. The selected technology needs to be the best match with adjacent steps before and after the capture step. For example, if you plan for a scenario with continuous perfusion culture as the source of feed for the capture step, the run time envisaged for the PCC step may be long, very different from processing the clarified supernatant of a fed batch reactor. This has consequences for the risk profile, for example, single-use technology as a precaution, mitigating risks for infection or contamination may be the logical choice to take. Dependent on scale of operation and system availability, other means of protecting the process have also been chosen, for example, steam-in-place capabilities on the skid. Looking for the best optimization route for such scenario, however, the option of daily capture processing of the pool collected from the perfusion reactor may yield similar overall process economy, with a small PCC or batch chromatography skid. To maintain the objective of a small footprint, the buffer supply to downstream processing could be designed using in-line conditioning (IC), essentially deleting buffer storage space requirements.

The facility to be used for the process may dictate technology choices, such as the selection of a single-use design, because the facility as a whole is designed to have no or limited CIP possibilities. Chapter 29 discusses the spread and design of single-use technology. At the time of this writing in 2017, there are a number of single-use facilities in operation, and more to come, but hybrid designs with single-use upstream and re-use concepts in downstream processing appear viable as

TABLE 28.10 Points to Consider When Comparing Batch and Continuous Mode Capture Methods

Aspect	Description	Comment
Production scenario	Feed from perfusion vs. fed batch	Differing load concentration Differing time constraints Vastly differing run time Differing risk profile and mitigation strategy
	Product demand	Single-batch vs. multiple batch Resin lifetime use in batch mode
	Facility design, facility fit	Fit with existing facility (su or ss) Open for new facility
Scale of operation	Adjust scale and cost to demand	Variability: max expected output Bioreactor and column scale
Resin consumption	Cycle vs. time productivity	Allowed processing time in batch and continuous mode scenario Cost of prepacked resin Available column dimensions
Buffer consumption	Criticality of volume and cost	In-line conditioning Realistic cost assumptions on buffer
Optimization	Comparable degree of optimization	Standard nonoptimized batch mode in case studies Limits of capability not explored Mass transport in turnaround steps

well (Amgen, Singapore). Very large scale stainless steel facilities are still being built to meet the maximum demand expectations for certain molecules in development. Dependent on the facility size and design, even in a new construction project, the savings to be expected from a single step may be considered marginal and technical advantages may get “ignored” or “steamrolled” by the overall facility design concept and its financial dimensions.

We still see case studies with comparisons of batch and continuous purification where the design of the batch process is limited to copying standard parameters with little optimization effort. Prolonged processing time for PCC is allowed, but not also used for the batch process. Improved utilization of resin lifetime is presented to be a benefit reserved for continuous processing independent on final manufacturing demand. Extra costs, for example, for single-use component consumption or prepacked columns are not included or not made transparent. And resin volumes compared are based on ideal calculations rather than on realistic use of available column sizes. Feed volumes and concentrations are “artificial” to the case study and the outcome may look different under different constraints for these parameters [8]. A flag should be on the table to make sure any model scenario is reviewed and verified for these aspects.

28.7.3 Example Calculations

The reference list for this chapter contains a large number of papers where model calculations on the benefits of PCC capture have been presented and discussed. In this paragraph, we are adding a discussion to illustrate some of the aspects discussed earlier: are there any other options to reduce the scale of the downstream process than switching to PCC mode? What impact should be estimated from extra costs for single-use or prepacked devices and from nonideal column dimensioning? How does a large-scale, regular manufacturing scenario compare to the early manufacturing for testing and trials under R&D regime?

We do not include aspects of facility cost, labor cost, or other parameters that are typically very specific to the user's scenario and rarely verifiable public knowledge. The case study is limited to verify resin cost savings and evaluate scenario dependent related cost at different product demand scenarios.

Model Calculations

The calculations are done using the author's model that allows to set a wide range of production scenarios as input to the batch or PCC mode capture step. that is, from very small demand and low titer to ton-scale demand and extreme product

concentration, for example, following a concentrated fed-batch cell culture. The model calculates the number of upstream batches to produce the demand and checks reality versus the available number and size of reactors, batch time for the reactor, and the available operating days in a year. Typical yields (95%) and dilution (10%) in the recovery part of the process are used to adjust titer information from the bioreactor to give load concentration and volume for capture. It is assumed that the processing time is limited, and for the purpose here an upstream batch has to be processed in no more than 48 h. A choice can be made on whether to allow the same or different processing times for both batch chromatography and PCC. Optimization is possible for the load and the turnaround cycle both on residence times, flow rates, and volumes per process step. Pressure limitations are implementable, and a selection of resins and column dimensions can be made. However, for this purpose we limit the choice to one of the high performing candidate resins for both batch mode and PCC capture of mAbs, MabSelect SuRe LX. Breakthrough and dynamic binding capacity data is used from internal reports at GE Healthcare.

We are using a slightly optimized batch process design as the basis. In particular, elution occurs in two column volumes instead of five recommended for many other resins. This is in line with discussions on manufacturing constraints related to the tank size for product intermediate and seems to be in line with industry optimization of the Protein A step. Other volumes and CIP time (15 min.) in the turnaround cycle are kept as often recommended, but can be adjusted. The postload wash is run at the same residence time as the load, but all other steps are accelerated to half the residence time, still clearly below pressure limit, which for the selected resin is 1000 cm/h at 10 cm (500 cm/h at 20 cm) bed height. Bed height is adjusted in iterations to find the best column and avoid packing a large excess of resin relative to the calculated need.

Single-Batch Scenario

In [Table 28.3](#), estimates are provided for product quantities supporting preclinical and early stage clinical testing, 0.5 and 4.0 kg, respectively. Assuming a titer of 5 g/L and a load concentration for the capture step of 4.32 g/L, a quick calculation shows that the former can be produced from one 200 L batch, and the latter from one 2000 L batch (78% DSP yield).

Selecting these reactors upstream, overproduction occurs: 240 g versus the preclinical demand and 3.4 kg for the early clinical case. In order to make this calculation demand driven, we assume that the large excess is not acceptable and run the batch at a turndown rate of the 2000 L bioreactor of 3/5, that is, at 1200 L reducing the overproduction to 450 g. We consider this also because the additional resin cost to support the additional 3.4 kg per batch would mask the expected demand-based resin saving from PCC completely. Note that the scale-down represents a reduction of resin use by 40% independent on which purification approach is taken. This is a large portion of what would otherwise be achieved by using the full bioreactor volume and a PCC capture approach. Scaling down to demand is the most powerful cost reduction strategy upstream when high cell culture productivity is available.

For the preclinical scenario, we accept the excess in support of other lab demand at this early stage. We test 24 and 48 h allowed processing time (equal for batch and PCC mode).

Load on the 4C-PCC system is performed at 2.4 min. residence time, which results in a load capacity of 59 g/L at 50% breakthrough. A fresh column is always available 17 min. before it is needed, the feed continuity constraint is met. DBC₁₀ at 6 min' residence time is 60 g/L. The load in the batch process was performed with a 20% safety factor, that is, load capacity was 48 g/L. Those load settings and the conditions for the turnaround cycle allowed to run 14 cycles (56 column loads) with PCC and 10 cycles on the batch column leading to the following column dimensions.

Preclinical Testing Supply

We have chosen to dimension the batch column with a similar very short bed height (5 cm) as for PCC (6 cm) to enable as many cycles as possible even for the batch run. With the small dimensions required here, that is still considered a feasible “pancake” format with few risks for robustness issues in the packing. Columns were matched to the calculated resin volume ([Table 28.11](#)) as much as possible without deviating too much from the target bed heights. The resin volume in 4C-PCC is only 23% lower than in batch. The purchase of this small quantity of resin in four prepacked columns is about \$10,000 US dollars more expensive than for the bulk quantity required to pack the batch column (we assume that the user can achieve average discounted purchase prices for both solutions). The total resin consumption is in the low \$10,000 US dollars cost range and might in fact not give rise to much of a cost discussion anyhow. Convenience of the prepacked solution for PCC will likely prevail in this case, but there are no savings to be expected versus bulk resin purchase.

Early Clinical Trial Supply

We matched the column dimensions to a 30 cm AxiChrom column with 17-cm bed height for the batch process and to a 14-cm prepacked column with 12-cm bed height (customized version) for PCC. This represents a PCC resin volume saving

TABLE 28.11 Results for Preclinical and Early Clinical Stage Manufacturing Scenarios; One Batch of 200 and 1,200 L, Respectively

	Resin Vol (L) Calculated	Resin Vol (L) Available	Extra Resin (%)	Resin Format
Preclinical supply (0.5 + 0.24 kg)				
Batch-1 (24 h)	1.52	1.57	3.2	Bulk
PCC-1 (24 h), per column	0.28	0.30	7.1	Prepacked
Early clinical supply (4.0 + 0.45 kg)				
Batch (24 h)	11.88	12.02	1.2	Bulk
PCC-1 (24 h), per column	1.73	1.85	6.9	Prepacked
PCC-2 (48 h), per column	0.84	0.86	2.3	Prepacked
Resin volumes calculated by the author's model: extra resin describes the difference between the model calculation and the actually available column sizes. Resin format determines the price level per L of resin.				

of 39% vs. batch (about 3% less than the ideal calculated value). To get to a close match vs the calculated resin volumes, initial bed height assumptions needed to be adjusted within limits. Allowing to extend the total process time for PCC to 48 h (29 cycles, 115 column loads), but not for the batch process, increases the resin volume reduction to 71% with prepacked columns of 10-cm diameter and 11-cm bed height. Facility time allocations including labor would go up proportionally, however, in this case. Allowing the same time extension even for the batch process returns the resin saving to the initial value at equal processing time. Whether it is realistic to allow a longer processing time or not for continuous processing remains case dependent.

The cost savings for resin at 24 h' processing time with PCC are estimated to 23% (instead of 39%) and at 48 h' processing time, to 53% (instead of 71%). The format cost for small prepacked columns reduces the advantage, relative to the reduction in resin volume (resin compression during packing is included, additional cost for custom column dimensioning is not included). These calculations do not consider the cost for disposable flow paths (if used).

We assume that the resin is for one-time use, that is, dedicated to manufacturing in one batch of the material for early clinical testing. Resins are dedicated to one molecule. While it may be difficult to re-use resin in prepacked columns, in the experience of the author the bulk resin does get re-used, for example, in laboratories (even for other molecules), or in later production of Phase III trial material of the same molecule. Typically, the shelf-life of the resin would allow doing this, if there are no other, overriding considerations. Doing this would wipe out the resin cost savings even under the most optimistic assumptions in favor of PCC, as long as no re-usable column format with bulk resin is applied even for continuous processing.

Keeping the numbers in mind, the cost argument needs careful checking. For example, the resin of the batch column would cost \$40,000–\$50,000 US dollars more than the four prepacked columns in the 4 kg early-stage trial scenario. The savings may be largely compensated for by the additional single-use component cost (one set per batch), an aspect that, by the way, does not speak against single-use in PCC (labor and facility time for packing for columns manually would be the alternative), but needs to be considered, if cost savings alone support the discussion in favor of PCC at this early stage of a development. There is not much left of the cost savings once all consumables are included in a relatively small single-batch scenario. Seemingly significant percentage values may distract from the fact that the dollar numbers are still small.

Multiple-Batch Scenario

We study a range of different product demands from 10 to 500 kg/year, which need to be met in a facility containing four 2000 L bioreactors. This configuration allows 80 batches to be processed assuming an upstream facility utilization of 80%. At a product titer of 5 g/L, each reactor delivers ≈ 17 batches for 500 kg, at a titer of 10 g/L it would be ≈ 9 batches per reactor and year (rounded numbers). With the same load residence times and load capacity as well as allowed processing time (24 h.) as for the early testing stage single batch scenario discussed earlier, required resin volumes for batch and PCC operation are as provided in Table 28.12. Column dimensions have been adjusted, as before, for minimal excess resin demand with industry typical fine tuning of bed height (23 cm for batch, 6 cm for PCC). The resin volume reduction with PCC is

TABLE 28.12 Resin Requirements in a Multibatch Scenario at 5 and 10 g/L Product Titer

Market Supply (10–500 kg)	Resin Vol (L) Batch	Resin Vol (L) PCC	PCC Vol (%)	Cycles Batch/PCC
Product titer 5 g/L	22.62	11.78	52	9/14
Product titer 10 g/L	36.58	11.78	32	11/28

approximately 50% and 60% at 5 and 10 g/L, respectively. To produce 10 kg at the low end of the range one needs to run 2 batches, or just 1 batch at 10 g/L of titer. Producing a demand of 500 kg requires to run 68 batches, or 34 at the high titer. We have assumed a resin lifetime of 160 cycles, which is valid for the batch scenario with full cleaning capability over this number of runs. For PCC where the column is a prepacked disposable column, the number of allowed CIP cycles and long-term bed stability is not publically documented. However, if we assume that up to 160 cycles are possible, the lower titer PCC process consumes the resin after loading of approximately 102 kg antibody, the batch process consumes the resin after loading of 158 kg (with the column configurations selected).

Calculated per liter of resin, the batch process can process 7.0 kg of mAb and the PCC process loads 8.7 kg. This is what the dynamic binding capacity of the resin under the selected conditions predicts. Once the resin is used for a more and more significant part of its lifetime, the advantage of PCC regarding better lifetime utilization melts away and the improvement in economics is entirely based on the better capacity utilization with PCC.

The difference in resin purchase per gram between batch and PCC mode as described before is shown for various product demand scenarios ranging from 10 kg to 500 kg annually (Fig. 28.17). We studied a facility with 2000 L production reactors. Specific costs per gram of antibody (A) are more favorable at higher product output and the savings decline exponentially between 10 kg and 100 kg demand. Between 100 kg and 500 kg annual output, the model shows savings between \$0.50 and \$1.50/g. PCC will always deliver a savings of resin volume based on better capacity utilization. In our scenario, the annual savings were found to be between \$200,000 and \$500,000 US dollars (B). The savings pattern with increasing facility output is irregular, since the PCC column replacement happens at a different time (earlier) than the refill for the batch operation. With increasing titer, the savings are generally higher. The calculations include resin compression during column packing for the batch mode column and the format cost for prepacked columns used in PCC.

An estimate of the column packing cost for each column refill shows a maximum of \$5000 assuming 8 h' packing time with two operators and a fully loaded hourly labor cost of \$250. These costs do not change the picture in any significant way. If a single-use flow path would be used on the PCC system, replacement costs would need to be deducted from the savings in the step. There is no consolidated and agreed experience for this yet. We have assumed that one complete flow kit for this large-scale operation could cost \$10,000 and would be exchanged after each bioreactor batch (*dotted lines* in Fig. 28.17B). This is currently a nonverifiable estimate, but it allows to make a quick calculation of the single-use costs even when the numbers change (costs increase or decrease proportionally with the kit price and the sequence of its exchange in the system) and provides a feeling for how much of the resin savings may be deleted by other expenses, for example, single-use component purchases.

Differences in buffer consumption have not been included in the calculation. However, a quick estimate for this case study reveals that the larger batch column may consume up to 8000 L of different buffers per batch. At an average buffer cost of \$0.50/L this would be a cost of \$4000 and the savings in PCC mode would likely be around \$1000 per batch. This may be considered insignificant in the context of the consumables costs and savings described before.

When testing for the effect a scale-down of the bioreactor for low demand scenarios would have on resin savings, we found that the total resin costs are obviously much lower (smaller columns for less load) and consequently also the savings are reduced (B, *yellow-gray*). This is in line with the single-batch scenario.

28.7.4 Summary—Putting Things in Perspective

Periodic counter-current chromatography (PCC) allows one to utilize the capacity of Protein A affinity capture resins better than batch operation. In situations where a limited product demand is produced, the technology also allows minimizing the resin volume further by running a higher number of cycles in the available time and thus reducing the column size. When the resin lifetime can be fully utilized both in batch and continuous operation, capacity utilization remains as the most significant benefit of PCC. The resin volume reduction percentage should not be equated with the cost reduction effect.

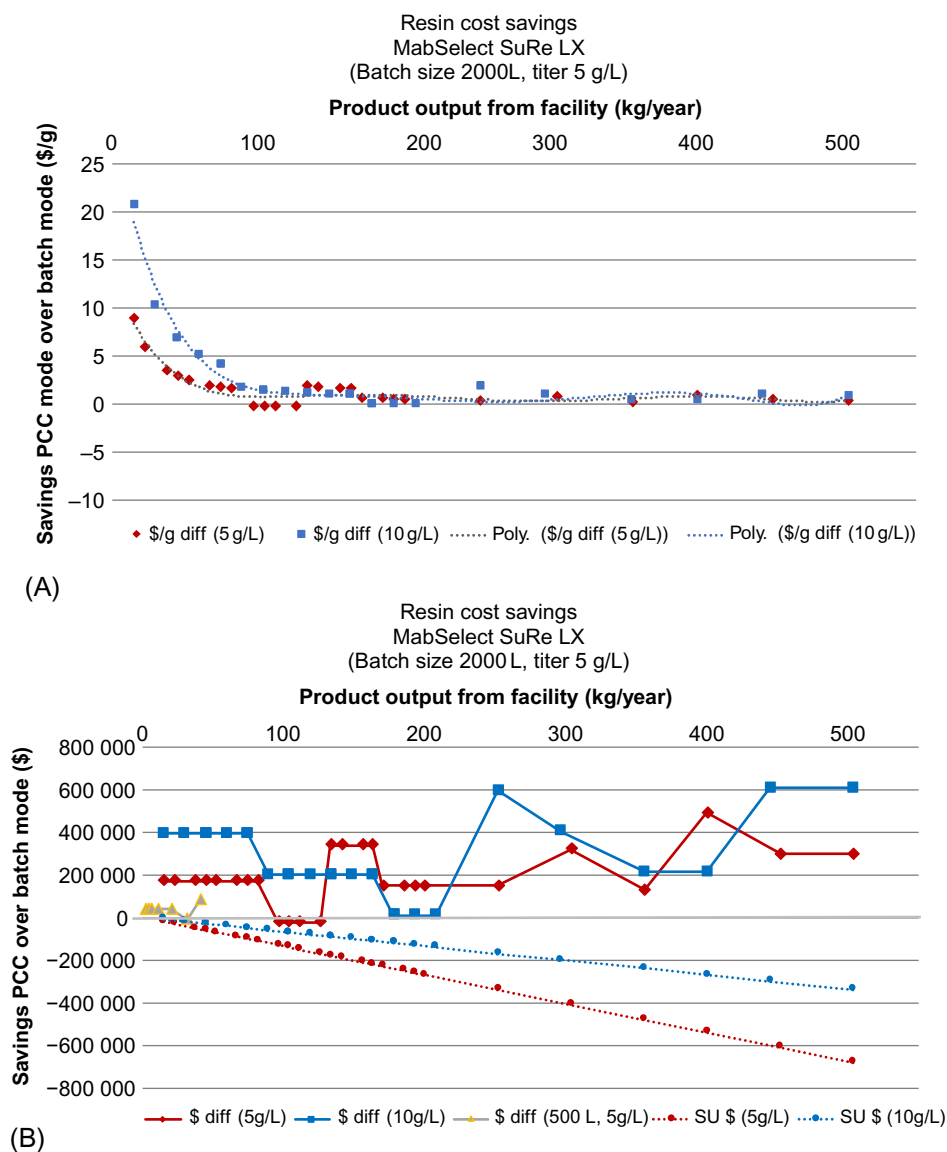


FIG. 28.17 Consumables cost reduction, comparison between batch mode and PCC mode affinity capture chromatography for monoclonal antibodies at facility output of 10–500 kg; (A) cost savings per gram of mAb at 5 (red) and 10 g/L (blue) product titer and 2000L batch volume; (B) cost savings in US dollars (solid lines, color code as in A), cost savings for low demand (1–40 kg) in reduced reactor volume (500L, yellow), potential cost of single-use flow path (dotted lines) for illustration (assumption: exchange every batch, cost per kit, \$10,000); cost savings show an irregular pattern due to different column refill/replacement schedule for batch and PCC mode setup.

Prepacked column format costs may delete the resin savings entirely in low demand scenarios and do reduce it under all circumstances. Still, prepacked columns will be the preferred format for PCC to avoid multiple column packing. Single-use flow path costs for PCC or SMB systems and the routines for more or less regular exchange of those flow paths are not yet established in the industry. In multiple batch scenarios, we estimate that these costs could reach the same order of magnitude as the resin savings as long as the “single-use” character of these flow paths is not modified towards longer term re-use.

28.8 ACKNOWLEDGMENTS

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